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Predictive Modelling of Flight Crash Risk- A Data-driven approach
using Multivariate Analysis and Machine Learning

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Submitted in partial fulfilment of the requirements for the Degree of Master of Science in Data Science

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Abstract

This study aims to develop a predictive model for diagnosing the risk of flight crashes using Machine learning techniques and Multivariate statistical analysis. The research involves integrating numerous collective datasets with well-defined features such as Mechanical description, weather conditions, Crew, Aircraft, Findings, and other operational variables that are relevant to aircraft crash scenarios into a single complete dataset suitable for analysis.

Using Exploratory Data Analysis, major determinants are recognised, the quality of the data is inspected, consistency is ensured, and the hidden patterns and interrelationships within the data are uncovered. The research applies Multivariate Statistical Analysis to investigate the complex interactions between different components and their contribution to flight crash risk, which leads to a deep understanding of the features. Using KNN, SVM, Logistic Regression, and Naive Bayes, we predict flight crashes by analysing whether the current flight features lead to crashes based on the early warned occurrence code.

The derived model gives a critical understanding of the sophisticated dynamics of aircraft crashes, providing a tool for early warning and risk mitigation. The results highlight how crucial it is to apply cutting-edge analytical techniques and integrate data to improve aviation safety. Further research will focus on developing the model, assimilating real-time data, and developing advanced prediction approaches.

Table of Contents

Abstract.....	1
1. Introduction:	9
1.1 Introduction to Aviation Accidents.....	9
1.2 Justification for research focus	12
1.3 Research Aim and Objectives	13
1.3.1 Research Aim	13
1.3.2 Research Objectives.....	14
1.4 Outline of the Research chapters.....	14
2. Literature review.....	15
2.1 Aviation accidents:.....	15
2.2 Insights of Multivariate statistical analysis	16
2.3 Machine learning in Flight crash prediction	18
2.4 Research Gap	21
3. Methodology:	23
3.1 Research Questions:.....	23
3.2 Research Objectives:	23
3.3 Research Design:	24
3. 4 Dataset:.....	25
3.4.1 Selecting the features and Justification:	25
3.4.2 Dataset description:	27
3.5 Data Pre-processing and Exploratory Data Analysis:	28
3.6 Multivariate Analysis for Predicting Flight Crash	29
3.7 Building a Predictive modelling for flight crash risk:	31
3.7.1 Generating synthetic normal data.....	31
3.7.2 Train and Test Data Split	32
3.7.3 Fine-tuning and Hyper-parameter.....	32
3.7.4 Predictive modelling algorithms	33
3.8 Justification for the Chosen Research Method	35
3.9 Alternative Research Methods for Predicting Flight Crash Risk	36
4. Analysis and Findings	37
4.1 Data Pre-processing & Exploratory Data Analysis	37
4.1.2 Detection and Correction of duplicate values	37
4.1.2 Handling missing values.....	41

4.1.3 Outliers detection and removal	44
4.1.3 Insights of the Dataset.....	46
4.2 Multivariate Analysis	50
4.2.1 Correlation Analysis	50
4.2.2 Principal Component Analysis (PCA).....	52
4.2.3 Factor Analysis	55
4.2.4 Cluster Analysis	58
4.2.5 Conclusion of the use of Multivariate analysis in predictive modelling	61
4.3 Predictive models for flight crash risk.....	63
4.3.1 Detailed Framework for Model Training and Evaluation	63
4.3.2 Performance of the models.....	65
5. Project management	73
5.1 Effective Time Management and Task Insights in Project Execution	73
5.2 Risk management:.....	74
5.3 Resource management	75
5.4 Quality management.....	75
5.5 Social, Ethical, Legal and Professional consideration	75
6. Achievements and Critical Evaluation of the Research	76
6.1 Key takeaways from the Analysis	76
6.2 Extent of Analysis in Answering the Research Questions and Objectives.....	76
a) Effectiveness of Supervised Machine Learning Algorithms:.....	76
b) Implications of multivariate analysis techniques:.....	76
6.3 Addressing the Research Objectives	77
a) Measuring and Comparing Machine Learning Algorithms:	77
b) Impact of Pre-processing Techniques:.....	77
c) Application of Multivariate Analysis Techniques:.....	77
6.4 Limitations of the Research	77
7. Conclusion	78
7.1 Achievement.....	78
8. Future Works	78
9. Student Reflection	80
References	81
Appendix	87

Table of Figures

Figure 1 Chart depicting the accidents and their causes throughout the years (Aksen et al., 2024)....	10
Figure 2 Predictive model development (Passarella et al., 2024)	20
Figure 3 Research design	24
Figure 4 Block diagram explaining label encoding Nishoak Kosaraju et al. (2023).....	28
Figure 5 Flow chart explaining working of KNN (Akhiladevi M et al., 2022).....	33
Figure 6 Scatter plot to detect noise	37
Figure 7 Computing statistical measures in numerical values.....	38
Figure 8 Computing statistics for each column after reducing the repeated values.....	40
Figure 9 Number of missing values in each column	41
Figure 10 Z-score for each column.....	44
Figure 11 Outliers in each column	45
Figure 12 Z-Score after removing the outliers	46
Figure 13 Wind speed by Occurrence.....	46
Figure 14 Visibility by light conditions	47
Figure 15 Yearly incident up to 2023	47
Figure 16 Age distribution for each crew category.....	48
Figure 17 t-SNE visualization of Feature Data	49
Figure 18 : Correlation Matrix of Features.....	51
Figure 19 PCA of the data	52
Figure 20 Cumulative explained variance by PCA components.....	53
Figure 21 Loadings of PCA components.....	53
Figure 22 Cumulative Explained Variance by Factors	56
Figure 23 Eigen value for each Factor	56
Figure 24 Dendrogram for Hierarchical clustering.....	58
Figure 25 K-means Clustering	58
Figure 26 Cluster centroids	59
Figure 27 Silhouette scores for Different numbers of Clusters	60
Figure 28 KNN- ROC curve	66
Figure 29 KNN confusion matrix	67
Figure 30 SVM – ROC curve	68
Figure 31 SVM confusion matrix	68
Figure 32 Naïve Bayes ROC curve	69
Figure 33 Naïve Bayes Confusion matrix	70
Figure 34 ROC – Logistic regression	71
Figure 35 Logistic Regression Confusion matrix	71
Figure 36 Accuracy comparison between the models.....	72
Figure 37 Gantt chart for research plan.....	73

Table of Tables

Table 1 Outline of the Research Chapters	14
Table 2 Dataset description	27
Table 3 Crew category code explanation.....	48
Table 4 Factor loadings	55
Table 5 Overall performance of the models	65
Table 6 KNN results.....	66
Table 7 SVM results.....	67
Table 8 Naive Bayes results.....	69
Table 9 Logistic regression results	70
Table 10 Risk management and mitigation	74

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1. Introduction:

1.1 Introduction to Aviation Accidents

In the transportation sector, aviation safety remains a paramount concern. Aircraft crashes pose a devastating loss of life, affect passengers and crew members, and have significant consequences for the airline economy. Predicting aircraft crashes before they occur helps in assigning authorities more productively for emergency management, reducing the impact on both human lives and aviation operations.

A registry review revealed more than 700 aviation-related deaths per year, although only 7% of those are related to commercial airliners. In communities without an airport, there remains the potential for an aircraft flying overhead to either crash or make an emergency landing with little warning and the potential for significant injuries to passengers and the local populace. (Sarin & Pruitt, 2024)

Complex interactions between several variables, such as sabotage, weather, mechanical failure, and human error, can lead to aviation accidents. To improve overall aviation safety and create effective preventive measures, it is imperative to comprehend these causes.

According to the International Air Transportation Association (IATA) (2021 Safety Report Edition 58. (2022), the data on Aviation accidents shows that numerous aircraft accidents were caused by unstable aircraft handling, erratic approaches, and unexpected weather penetration. Which is the largest percentage of causative factors (37%) in aircraft handling. 34% of accidents are caused by environmental threats. Runway excursion has the highest number of accidents in the accident category. (Lee & Truong, 2024)

Accidents were more commonly attributed to mechanical/technical and pilotage reasons during the period from 1941 to 1950. Additionally, the occurrences of meteorological and unknown reasons were also higher during this time frame. Subsequently, from 1951 to 1980, the distribution of reasons for accidents was relatively close, with mechanical and pilotage reasons accounting for accidents from 1941 to 1970 and 1971 to 2000, respectively. The percentages of accidents for mechanical and pilotage - 51.60%, meteorological reasons - 28.76%, and unknown - 19.64% respectively. (Aksen et al., 2024)

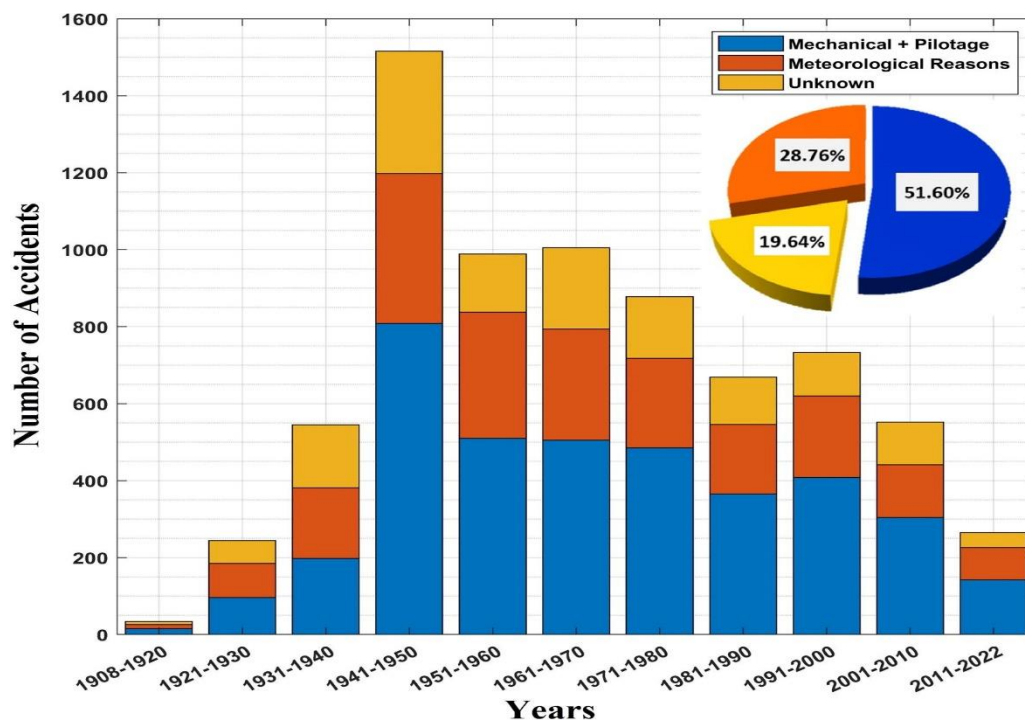


Figure 1 Chart depicting the accidents and their causes throughout the years (Aksen et al., 2024)

Aviation Safety and Prevention measurements:

The motivation behind studying aviation accident reports is grounded in the aviation industry's unwavering commitment to safety enhancement. This commitment has led to significant advancements in technology, training programs, and regulatory measures. However, accidents persist, often resulting from intricate interplays between human factors, technical failures, and environmental conditions. Timely and precise analysis of accident reports is imperative for preventing future incidents and elevating aviation safety measures. (Aziida Nanyonga et al., 2023)

Accidents still happen because of intricate interactions between environmental factors, technical malfunctions, and human factors, even with these advancements. Thus, improving aviation safety protocols and averting future mishaps depends on the prompt and accurate analysis of accident reports.

To ensure aircraft operational safety, most countries established a relatively unified safety management system and risk assessment measures. At the national level, each country has established an incident and accident database and in-flight medical incapacity database to control the overall safety of air transport and implemented national aviation reviews and state safety programs. The mechanics of risk management are gradually shifting from a traditional accident investigation to a data-driven approach. Of course, both risk management approaches now exist simultaneously. (Guo et al., 2022)

Studying aviation accident reports is motivated by the industry's dedication to improving safety, which has resulted in developments in training curricula, technology, and regulations.

After developments over decades, the civil aviation industry has accumulated abundant safety data sources, including information on various types of safety events like accidents, accident proneness, and other safety events, monitoring information on flight quality of airline companies, industrial supervision information, evaluation information of maturity of the safety management system, and so on. (Sun et al., 2023)

Challenges and Risks in Analysing Aviation related data:

Analysing aviation accident data bears several challenges to affect the accuracy and reliability of their results. The majority of the data are unstructured; hence, extraction of relevant details does not always have consistency. The amount and quality of data and inconsistencies, incomplete data, and volume result in big challenges. The study of human factors remains complex, and high classification accuracy cannot be easily achieved as huge labeled datasets are needed during training.

Accident analysis is carried out to determine the causes of accidents and prevent further recurrences. Accident and incident records provide one way for learning; however, these are usually unstructured texts, making analysis difficult. Various text mining methods based on supervised learning have recently been proposed for analysing accident reports; however, they often require a large, labelled training dataset. Most of the semi-supervised and unsupervised approaches that seek to overcome this shortcoming suffer from inadequate classification accuracy and limitations in cross-domain applicability. (Ahadh et al., 2021)

The consideration of dataset size and the deployment of advanced techniques need to be considered for innovation in aviation safety. Future work will look at the impact of combining the ASN and NTSB datasets. (Aziida Nanyonga & Wild, 2023)

Several problems arise when carrying out an aviation accident analysis based on a large, complex dataset such as the NTSB data set, which is used in this research. When the dataset size is large, risks of incomplete or inconsistent entries increase, which further degrades the reliability of such analysis. Much of the effort for cleaning and structuring data for analysis would get soaked into this humongous dataset. Structural text entries may make it hard to derive meaningful insights from Accident Reports. Added to these are varying report formats and terminologies that sometimes pad up all these facets, thus making the integration and standardization of data across records even more difficult.

1.2 Justification for research focus

- I. **Enhancing Aviation Safety:** Given the high stakes and complexity of air travel, there must be improvement in aviation safety. For this, accident reports can be evaluated to understand the rationale behind the incidents and help in avoiding such mishaps in the future.
- II. **Huge sources of data:** Accident reports, data related to flight quality monitoring, safety management evaluations—these are huge amounts of data generated in the aviation industry. However, much of this data is unstructured, posing significant challenges for traditional analytical methods. Acres of unstructured data are a product of the aviation industry, and accident reports, flight quality monitoring data, and safety management evaluations have considerably challenged traditional analytical methods.
- III. **Limitations of Traditional Methods:** Traditional methods of accident analysis cannot often deal with unstructured text data and also require large and labelled datasets for supervised learning. While semi-supervised and unsupervised approaches have insufficient classification accuracy and limited cross-domain applicability. Semi-supervised and unsupervised approaches demonstrated rather low accuracy of classification and poor cross-domain applicability.

- IV. **Advanced analytical techniques:** Traditional procedures can be overcome by using advanced data analysis techniques like statistical methods and machine learning algorithms. These methods improve categorization precision and offer a more in-depth understanding of the causes of accidents.
- V. **Integration of Comprehensive Datasets:** One can have a more solid footing for analysis by making use of comprehensive datasets such as the NTSB dataset. This information would be very useful for improving data collection processes and thus working toward a better understanding of aviation safety. It is possible to set up a strong base for analysis through comprehensive data sets like NTSB datasets. It may also improve practices in data collection and eventually lead to the understanding of stringing aviation safety.
- VI. **Data Analysis Innovation:** Regarding several main characteristics of this dataset, utilizing state-of-the-art analytical techniques takes innovation in aviation safety to a new level, with the objective of identification methodologies that retain meaningful insights from it, bearing in mind the complications of structured data. This research is going to implement a difference in aviation safety by considering the size of the dataset and using advanced techniques of analysis. It is considered for developing methodologies that will be able to make use of the complexities of unstructured data and offer meaningful insight.

1.3 Research Aim and Objectives

1.3.1 Research Aim

This research aims to apply Machine learning and analyse various dimensions from large datasets, coming up with a model that can predict the risk associated with plane crashes. The research relies on multivariate statistical analysis and exploratory data analysis (EDA) to find the main causes of aircraft accidents. This will allow for the potential to build a strong predictive framework that will enhance safety precautions and aid in averting further catastrophes. The ultimate goal of the project is to improve airline safety and risk management procedures by transforming unstructured accident data into useful insights.

1.3.2 Research Objectives

- a) Using Exploratory Data Analysis methods to comprehend the distribution, patterns, and correlations within the datasets, perform in-depth exploratory data analysis, determine the essential elements and factors that have a major influence on the probability of aviation accidents, and ensure the quality of data by identifying missing values and outliers.
- b) Using machine learning and statistical methods to choose the most pertinent features to be used in the predictive modelling. To assist in identifying the elements contributing to aircraft crashes, extract significant insights from the data.
- c) Using Machine learning algorithms such as KNN, SVM, Logistic regression, Naïve Bayes to Create and hone machine learning models based on the chosen features to forecast the likelihood of flight crash possibility based on the early occurrence code warnings to predict the anomaly of risk of the crash.

1.4 Outline of the Research chapters

Table 1 Outline of the Research Chapters

Chapter No	Chapter content
1	Introduction to Aviation accident and backgrounds
2	Literature review on the related works
3	Methodology and research design along with the research approach
4	Analysis and Findings from the research
5	Achievement and Critical Evaluations
7	Conclusion
8	Future work and implementation
9	Student reflection

2. Literature review

2.1 Aviation accidents:

A registry review revealed more than 700 aviation-related deaths per year, although only 7% of those are related to commercial airliners. In communities without an airport, there remains potential for an aircraft flying overhead to either crash or make an emergency landing with little warning and the potential for significant injuries to passengers and the local populace. (Sarin & Pruitt, 2024)

As per (Kane et al., 2021), Human error, sabotage, weather, technical failure, and pilot mistake are the main causes of aircraft accidents. About 50% of accidents are caused by pilot error, and 20% are the result of mechanical issues. Even with the help of technology, weather-related problems persist and cause accidents. Human mistake and sabotage are also important factors. A 2000–2020 crash analysis identifies structural breakdowns, especially in engines, as the primary culprits. The relevance of engine pod sites is shown by an analysis of 111 crash reports, with typical locations falling between 20% and 40% of the wingspan. In order to reduce these complex dangers, this research emphasises the necessity of comprehensive safety measures and prediction models.

By addressing human error and system failures across the course of the aircraft's life cycle, aircraft systems design plays a critical role in increasing safety. This study integrates human factors analysis from the design phase, emphasising the necessity for robust, error-tolerant, and error-resistant systems. It draws attention to the probabilistic character of aircraft mishaps, which frequently stem from a confluence of variables rather than isolated incidents. The study also emphasises how crucial balanced automation and sophisticated fault diagnostics are to supporting pilot decision-making in emergency situations. The aviation sector may more effectively anticipate, prevent, and manage possible safety issues by implementing complete safety methodologies. Haider (2020)

2.2 Insights of Multivariate statistical analysis

The research (Lee et al., 2024) employed unsupervised learning approaches to identify aberrant circumstances in manufacturing processes, with a focus on multivariate data. When conducting preliminary data analysis, Principal Component Analysis (PCA) is utilised to efficiently identify anomalies. The study creates a comprehensive abnormality detection system known as the "Multivariate PCA-based Composite Criteria Evaluation Method".

This study (Ong et al., 2022) assesses how multivariate analysis and cross-validation techniques affect regression model performance. According to the findings, k-fold cross-validation usually produces better model performance, while results aren't always improved by higher k values. The right mix of reliable predictors is more important for multivariate analysis than the total number of variables when it comes to model efficacy. The significance of predictor selection and validation strategies in regression models is emphasised by this study, which also recommends more research into how these factors affect other machine learning methods.

The goal of the project S. Selva Kumar et al. (2023) is to use regression techniques to create a sophisticated restaurant recommendation system while addressing the dataset's intrinsic complexity. The results highlight how data exploration can have a transformative effect and show how regression analysis can reveal patterns and spur creativity. This study emphasises the value of data-driven solutions in contemporary society and shows how these approaches can promote more intelligent, data-driven decision-making in a variety of fields.

This study (Balaji Vs et al., 2024) emphasises the value of factor analysis in health research, particularly in relation to chronic liver disease. By looking at the loadings of each factor, which show how much each variable contributes to each component, factor analysis aids in the comprehension of the intricate interactions between the different qualities of the dataset. This procedure helps determine communalities and specific variances, as well as which variables are most closely related with each underlying element.

Regression analysis was used in the study (Anush Kumar Sivakumar et al., 2023) to find non-linear connections between variables, with a focus on crash dynamics and

quad-rotor descent under single motor and total propulsion failures. Crash radius predictions were given by quadratic regression models, together with prediction intervals that showed how unreliable these forecasts were. Prospective avenues for investigation encompass delving into failure modes of partial propulsion, examining supplementary crash dynamics elements, and verifying results through experimental investigations.

(Zou, 2022), this paper is an FT-based multi-state Bayesian network system security analysis model. The proposed framework integrates causal and diagnostic thinking to ensure a rigorous safety study from the qualitative to quantitative management. This polymorphic BN model would improve probability analysis, hence raising the scientific accuracy of safety evaluations. It can also be adapted for use at both the macro level of industry safety management and the micro level of airline operations. In the future, more data collection work will be conducted to make the model more practical in application, so as to become a reliable and efficient tool for civil aviation safety management. This research follows a comprehensive data-driven approach to solving safety problems in civil aviation, promoting the continuous development of safety management systems.

The reviewed material offers insightful analysis and useful approaches that will greatly advance my study of flight crash analysis. One way to find anomalous situations in flight data is to use PCA and unsupervised learning for anomaly identification in manufacturing processes. When validating prediction models in aviation accident situations, the literature review highlights the significance of k-fold cross-validation and predictor selection in regression models. Additionally, it emphasises the application of factor analysis and regression, respectively, which can aid in comprehending intricate relationships and variables influencing flight safety. Lastly, non-linear regression techniques and prediction intervals—which are essential for forecasting crash outcomes in aviation—are shown by the research on quad-rotor crash dynamics under failures. By including these insights, flight crash analysis procedures are improved, leading to improved safety evaluations and increased operational reliability in the aviation industry.

2.3 Machine learning in Flight crash prediction

Which causes and rates of aircraft crashes occur most frequently were covered in the publication "Aircraft crash rate and cause detection using machine learning techniques." To determine the exact result, they have used the Random Forest and Bayesian classification techniques. By using these machine learning techniques, they were able to determine the aircraft crash cause that occurred most frequently based on accident rates. According to their data, the most common causes of aircraft crashes include weather-related, bird strike, technical issues, and pilot mistake. Based on the dataset they employed Engine malfunctions and fuel starvation are the two most common causes of aeroplane crashes. (Kumari et al., 2023)

In order to predict the severity of aircraft accidents, a Random Forest and Decision Tree machine learning classifier are suggested in the study by M. Sunil Kumar et al. (2023). They employed logistic regression, a well-liked and intelligible method for binary classification applications since it provided insights into feature significance and probabilistic predictions. The data that we are using for our research is noisy and contains outliers. This work implies that individual decision trees are less sensitive to noisy data and outliers.

Three models were used to predict the outcomes of aviation accidents in the study by Prajwala T.R et al. (2023): XGBoost, Random-Forest Regressor, and Gradient-Boosting Regression. When it came to prediction accuracy, the Random-Forest Regressor performed comparably better than the others. The significance of data preprocessing—which is essential for enhancing model performance—was heavily emphasised. This emphasis on preprocessing and data quality highlights the study's all-encompassing strategy for enhancing aircraft safety via machine learning.

In the study (Mehta et al., 2021), By combining the performances of several contributing models, the Stacking Ensemble Model in this study maximises the accuracy of 91.66% while utilising the advantages of various models. Artificial Neural Networks (ANNs) demonstrated the highest accuracy of 91.51% across all individual models, while the Random Forest Classifier demonstrated the highest accuracy of 91.42%, indicating its capacity to handle a wide range of features. The results highlight the ability of ensemble models to improve prediction accuracy, offering important new information for advancing aviation safety protocols.

In order to determine whether or not flights are accident prone, Mathur et al. (2017) developed a model for aviation accidents in their publication. Usually, the model forecasts the circumstances in which an accident will occur. The suggested logistic model, whether tested using the same or different datasets, provides good prediction and recall values with increased accuracy. The estimation of structural damage brought on by flying bird strikes was the main goal of the Nimmagadda et al. (2020) paper. They took into account landing gear, engine attack, and aircraft damage. They employed machine learning methods like Naïve Bayes (87% accuracy), KNN (83% accuracy), and Decision trees (80% accuracy). They demonstrated that Gaussian naïve Bayes is the most efficient solution for this problem by running these algorithms with the identical inputs. In comparison to the current approach, performance increases in the same way.

The study (Passarella et al., 2024) compares and contrasts 25 prediction models for the identification of flight abnormalities using criteria such as accuracy, balanced accuracy, ROC-AUC, F1 score, and processing time. With a runtime of 2.46 seconds and an accuracy and balanced accuracy of 0.93, Quadratic Discriminant Analysis (QDA) was found to be the most effective model. In order to improve understanding of contributing aspects including weather, human error, and engine damage, the research emphasises the integration of statistics, machine learning, and artificial intelligence theories in model creation. The results provide insights into aviation safety and tactics for preventing anomalous flight patterns, with important theoretical and managerial ramifications.

While our dataset is large and has many columns, the (Rade Kačar et al., 2023) publication discusses dimensionality reduction, which is crucial to increase the performance of models. Principal Component Analysis (PCA), t-distributed stochastic neighbour embedding (t-SNE), and autoencoders are a few popular methods for reducing dimensionality. In the trained model, the boosted tree algorithms XGBoost and LightGBM yield better results.

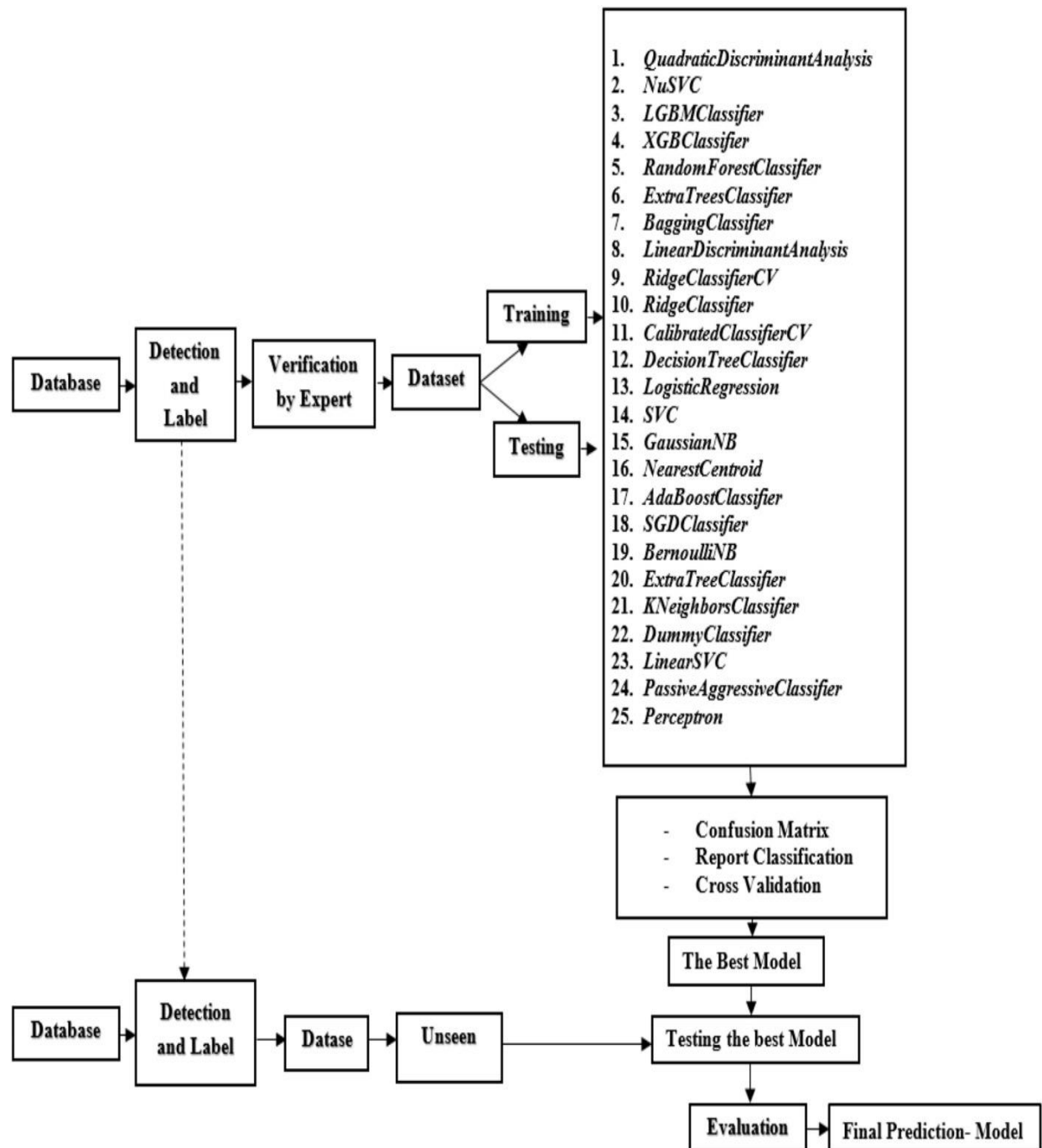


Figure 2 Predictive model development (Passarella et al., 2024)

This paper (Yunhan, 2023) has focused on the impactful strides gained in the area of aviation safety from the use of new machine learning aptitude through Aeronautical Safety Reporting System data analysis specifically. Ensemble classification models are widely used by researchers in fighting out the imbalanced of data involved in civil aviation safety reports. Previous studies emphasize that some models, such as Random Forest, XGBoost and Light Gradient Boosting Machine, help to increase the classification accuracy and speed. These models shall not only classify the situation of being unsafe, in general aviation, but also shall identify the causative factors of injury, thereby helping in timely bound interventions to reduce risk and hence shall improve the general aviation safety.

The paper Ved PrakashGupta (2020) acknowledges that no one algorithm consistently produces ideal results under all scenarios, but emphasises the usefulness of utilising various machine learning methods to increase forecast accuracy for aircraft safety. XGBoost leads the other four models in terms of performance: Neural Network, CHAID, Tree-AS, and XGBoost. These models are combined in the ensemble approach to reduce the shortcomings of each one and improve the accuracy as a whole. This multi-model approach greatly improves aviation accident severity estimates' predictive power and dependability.

Although such algorithms have already proved to be very effective in finding outliers and strange patterns, anomaly detection algorithms have received very little attention within machine learning literature. In this work, we will try to close this gap by applying anomaly detection techniques to the examination of aviation safety information. The accuracy and dependability of predictive models from these algorithms will improve, thus moving us further in predicting and reducing risks, hence raising the standard of safety for air operations.

2.4 Research Gap

Even though algorithms like KNN, SVM, Naive Bayes, and Logistic Regression are being extensively used, anomaly detection itself has not been given much attention in machine learning literature. Most of such algorithms predict anomalies blindly. In this paper, we focus on identifying risks based on occurrence codes to improve the prediction of crash probabilities, hence improving aviation safety.

Although machine learning (ML) is being used more and more in aviation safety, there is still a significant research gap in creating reliable ML models that are specifically designed to forecast and prevent aircraft crashes. Instead of real-time prediction capabilities, current machine learning applications frequently concentrate on retrospective analysis of crash data. Subsequent investigations ought to focus on improving machine learning algorithms to manage intricate aviation datasets in real-time, integrating elements like meteorological conditions, pilot conduct, mechanical malfunctions, and air traffic dynamics. Furthermore, there is still much to learn about anomaly detection methods that can foresee and stop unforeseen incidents in real-time aviation operations. Closing these gaps could result in improved aircraft safety procedures and stronger predictive models. In order to improve proactive safety measures and accident prevention tactics in aviation, approaches that can manage the multidimensional character of aviation data, including spatial and temporal aspects, are also required.

Interpretable machine learning models are also required in order to improve safety procedures and accident prevention tactics in the aviation sector by offering aviation stakeholders useful insights and decision support. Rather than emphasising predictive analytics, current statistical methods frequently concentrate on descriptive and inferential analysis of past data. In order to gain a deeper understanding of the intricate relationships and interdependencies that cause aviation accidents, future study should investigate the application of sophisticated statistical approaches such as Bayesian networks, hierarchical modelling, and spatial-temporal analysis.

There is still a gap in the comprehensive integration of regression, factor analysis, and anomaly detection approaches for proactive risk management in aviation, despite the literature study highlighting their application in numerous facets of aviation safety. In particular, research is required that combines factor analysis to pinpoint the crucial elements affecting safety outcomes with regression analysis to simulate nonlinear interactions in flight dynamics.

3. Methodology:

3.1 Research Questions:

- a) How effective are different supervised machine learning algorithms in predicting flight crash risk in aviation safety data?
- b) What is the impact of multivariate analysis techniques on improving the accuracy and interpretability of models predicting flight crash risk based on the occurrence code?

3.2 Research Objectives:

- a) To Measure and compare different supervised machine learning algorithms with respect to K-Nearest Neighbours, Support Vector Machine, Logistic Regression, Naive Bayes, for flight crash risk predictions.
- b) To Study how different feature scaling and encoding, among other pre-processing techniques, affect these machine learning models' performance in predicting flight crash risk.
- c) To apply multivariate analysis techniques, to enhance the accuracy and interpretability of models predicting flight crash risk based on the occurrence code.

3.3 Research Design:

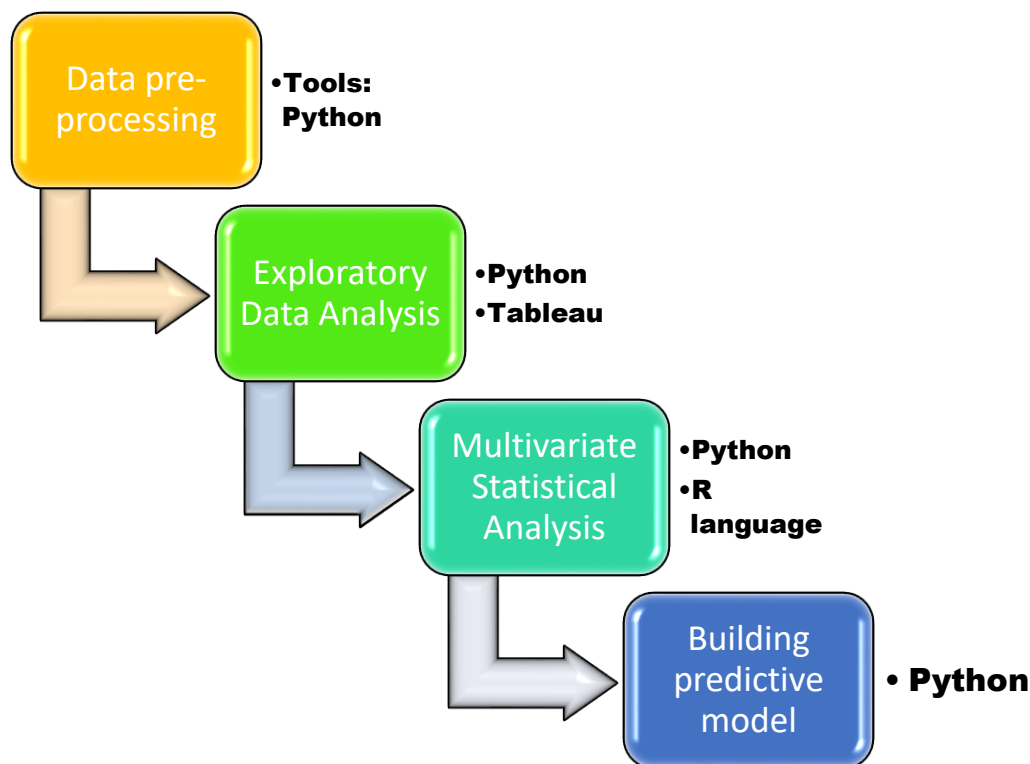


Figure 3 Research design

This research is attempting to provide a risk prediction of flight crashes using features derived from flights and readings after the issuance of an occurrence code alert. This research predicts flight crash risks using multivariate analysis with machine learning techniques. Data preprocessing was the first stage of this project, followed by encoding categorical variables, after which it had to standardize features. This is followed by exploratory data analysis methods—heatmaps, pair plots, and correlation matrices—to understand the distribution and relationships of data. Various machine learning algorithms have been applied, specifically KNN, SVM, Logistic Regression, Naive Bayes, for building predictive models. Train and test the models, then check model performance through metrics such as accuracy, precision, recall, and ROC curves. The results are visualized for comparing the actual vs. predicted outcomes, and the best model that would perform well in effectively predicting crash risks can be identified.

3. 4 Dataset:

This project uses the dataset provided by the National Transportation Safety Board, United States. The source and details about the dataset are available on the NTSB Data and Statistics webpage.

(https://www.nts.gov/safety/data/Pages/Data_Stats.aspx.)

Version of dataset used can be downloaded here.

<https://data.nts.gov/avdata/FileDirectory/DownloadFile?fileID=C%3A%5Cavdata%5Cavall.zip>

Timeframe : Years 2008 to present

There are sixteen different datasets in the zip files that can be downloaded. The datasets include flight details, occurrence details, event descriptions, weather conditions, crew details, and location details that encompass a huge amount of information related to aviation accidents. Additionally, a detailed dictionary is included to help interpret the codes and features within the datasets.

3.4.1 Selecting the features and Justification:

The zip archive includes almost 400 columns in 16 datasets, the majority of which consist of investigation details and analyses results obtained after an aviation crash. From the point of view of this study, only those features that may have contributed to the crash are relevant.

A meticulous feature selection process was conducted after the perusal of all columns against the comprehensive dictionary provided. This was done with due care and required extensive analysis in selecting only those columns of interest. To determine which columns were most pertinent, a thorough examination and thoughtful deliberation were required. The feature selection was a very important, confidential part of the research, done with the highest degree of care to include only those features that turned out to be most relevant.

The selected columns have been extracted from each dataset and compiled into a single CSV file using Python.

Occurrence_Code:

A unique identifier code for every single occurrence is provided, which becomes very important in tracking and analysing certain events. Occurrence categories are used to classify occurrences (that is, accidents and incidents) at a high level to permit analysis of the data in support of safety initiatives. Categories, such as CFIT and “loss of control” have been developed specifically for this purpose. *AVIATION OCCURRENCE CATEGORIES DEFINITIONS and USAGE NOTES* (2013). The Occurrence Code is crucial for our research because it serves as a primary indicator of risk. The detected Occurrence Code, along with other phases and features of the flight during the incident, will form the basis for risk prediction.

Aircraft details:

Adding aircraft models as a feature in a predictive model improves its accuracy, since aircraft differ in design, performance, maintenance, and historical safety records, number of engines, and engine type as features in predictive models improves accuracy by capturing critical design and performance differences, enhancing aviation risk assessment.

Crew details:

Involving crew category and crew age into predictive models is important because all these factors bear a direct impact on civil aviation safety. The crew category indicates different levels of experience and training. Crew age can be related to performance, physical condition and conforms to certain standards of health. Both these features contribute to the successful risk assessment and prediction of possible safety-related problems.

Flight phase:

In predictive models, parameters like longitude, latitude and altimeter readings are critical. These provide context and environmental conditions in real-time and may be used in the accurate assessment and prediction of risks in aviation by reflecting the position and flying condition of the plane at any given moment.

Environmental factors:

Light conditions, visibility, wind velocity, wind direction, temperature, and weather code are the factors that a predictive model should contain in order to capture the assessments of risks in regard to aviation; all of these environmental factors offer very important context toward the assessment and prediction of hazards bound to flight in order to ensure greater reliability and better decision-making.

3.4.2 Dataset description:*Table 2 Dataset description*

Column name	Description	Measuring values	Data type and unique values
Occurrence_Code	An identifier code	---	Numerical
acft_model	Aircraft model	---	---
num_eng	Number of Engines	---	Numerical Unique values: 1 and 2
eng_type	Engine type	---	Categorical ELEC (Electrical) REC (Reciprocating) TF (Turbo fan) TJ (Turbo jet) TP (Turbo prop) TS (Turbo shaft) UNK (Unknown)
crew_category	Pilot Category in command	---	Categorical (Captain, Co-pilot, First Pilot and etc.)
crew_age	Age of the crew	---	Numerical
latitude	Phase of the flight	Degree (°)	---
longitude	Phase of the flight	Degree (°)	---
light_cond	Light Condition	---	Categorical
wx_temp	Air Temperature	Degree Celsius	Numerical
wind_dir_deg	Wind direction	Degrees magnetic respect to North	Numerical
wind_vel_kts	Wind velocity	Knots	Numerical
altimeter	Pressure	Hg	Numerical
code	Sky condition	---	Categorical

3.5 Data Pre-processing and Exploratory Data Analysis:

1. Dataset Description: The primary dataset used in this work is an aviation safety dataset with several features that describe each flight, including weather and aeronautical information, among many others. The target variable in the case would then be the occurrence code that defines several kinds of flight incidents or crashes.

2. Data Cleaning: This would be the first step in cleaning up the dataset to assure there are no missing values, outliers, and inconsistencies for enhancing the quality and reliability of data for analysis. Due to the data's enormous and widespread nature, the selected columns contain a significant amount of missing values. Handling missing values and outliers is explicitly explained in Chapter 4.

3. Feature Encoding: The encoding of categorical variables, for instance, 'acft_model', 'light_cond', 'eng_type', 'crew_category', 'latitude', 'longitude', and 'code' variables, ensures that they are represented in a form suitable for machine learning algorithms through techniques like label encoding or one-hot encoding, which converts a categorical variable into a numeric form.

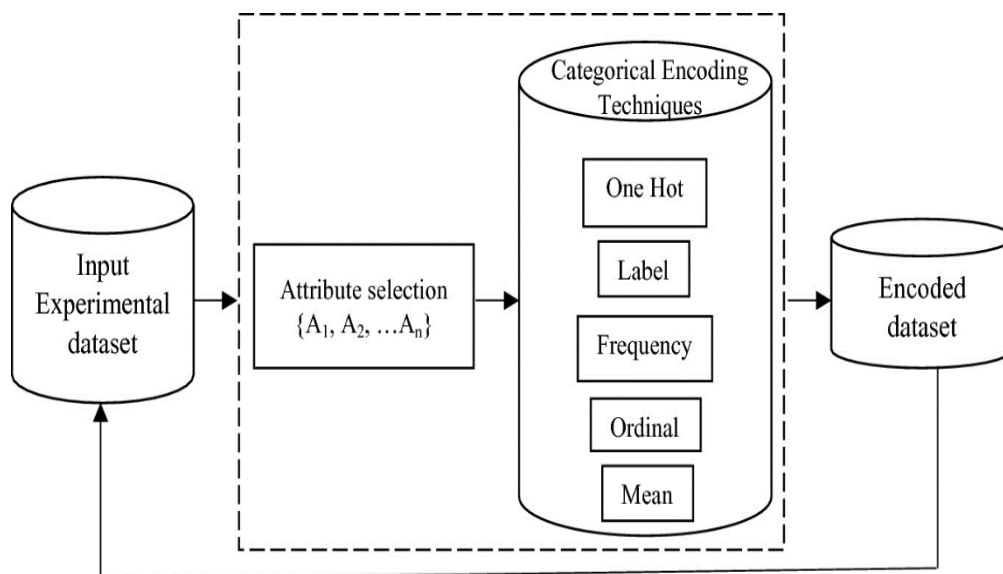


Figure 4 Block diagram explaining label encoding Nishoak Kosaraju et al. (2023)

4. Feature Scaling: Numerical features are standardized or normalized using algorithms like StandardScaler to ensure all features contribute equally towards the model performance. StandardScaler is sensitive to outliers, and the features may scale differently from each other in the presence of outliers (Scikit-Learn, 2019). This ensures that each feature is equally important to the model by normalizing data to some standard scale.

5. Ensure the Quality: Exploratory Data Analysis (EDA) helps ensure data quality by identifying missing values, detecting outliers, and uncovering data distribution patterns. It facilitates understanding data structure, validating assumptions, and revealing relationships between variables, which aids in cleaning and preparing the dataset for robust modelling.

3.6 Multivariate Analysis for Predicting Flight Crash

Multivariate analysis methods contribute to large improvements in the accuracy, interpretability, and efficiency of our model for predicting the risk of flight crashes. The list includes, among others, Principal Component Analysis, or the PCA, Correlation Analysis, Cluster Analysis, and Discriminant. These value-adding techniques refine the dataset and enhance the machine learning predictive models.

Principal Component Analysis:

Principal Component Analysis is a technique for dimensionality reduction. A set of correlated features are transformed into another feature space called principal components, in which these features are not correlated. The components capture the maximum variance of the data; the first few components account for the most significant portion of the total variance. We can apply PCA to reduce the dimensionality while keeping the important information about the dataset. It enables the construction of more interpretable models and reduces the risk of overfitting, especially on high-dimensional data.

Correlation analysis:

Correlation analysis is done by calculating the correlation coefficients of pairs of features, helping in understanding their relationships and dependencies. Highly correlated features will lead to redundancy and further multicollinearity in the model; this degrades its performance. The analysis will help in the elimination of redundant features but keeping relevant ones that will ensure both efficiency and accuracy for the model. This is therefore a very important step in preserving the integrity of the predictive model because it helps select the subset of features that provide unique and valuable information without raising the model complexity.

Cluster Analysis:

Cluster Analysis applies techniques like K-means or hierarchical clustering to group similar instances of data together based on their feature values. It is a very useful method for revealing the natural structure and patterns within a dataset. Interpreting these clusters shows us the different types of flight incidents and causes associated with them. Cluster Analysis is very helpful in identifying distinct patterns that otherwise would not be apparent from individual features alone. It helps to deepen our understanding of data, which may further be helpful in the development of more focused and effective predictive models.

Factor Analysis:

Factor analysis allows for simplification of complex data sets by the identification and grouping of latent variables in an underlying dataset, which capture all information from several measured variables. This technique is useful in dimensionality reduction since it becomes easy to interpret the data in terms of a few significant factors rather than lots of variables. It also accounts for multicollinearity by turning correlated variables into uncorrelated factors, hence increasing the reliability of subsequent analyses. Factor analysis, at its simplest, serves to slim down the data, uncover hidden patterns, and make sure that the models are based only on those features which have the most impact, hence leading to more accurate and meaningful insights.

In other words, applying principal component analysis, correlation analysis, cluster analysis, and discriminant analysis in our research will help in dataset refining, improving the performance of the models, and making appropriate predictions for the risk of a flight crash. Each method adds value to the knowledge of data and generally increases the quality and efficiency of predictive models. These multivariate techniques will allow us to develop models that are strong, enabling the production of reliable risk assessments to enhance safety and reliability in flight operations.

3.7 Building a Predictive modelling for flight crash risk:

3.7.1 Generating synthetic normal data

Synthetic data is data that has been generated using a purpose-built mathematical model or algorithm, with the aim of solving a (set of) data science task(s). (Jordon et al., n.d.)

Synthetic normal data provide a baseline of what constitutes "normal" in the context of data. In training and validating anomaly detection algorithms, this is very critical. Normally, such algorithms require contrasting normal and anomalous patterns. It helps to evaluate how well the model performs in anomaly detection. Generation of synthetic normal data would help in sampling through such scenarios and ensure that the anomaly detection system is tested and fine-tuned properly, even if there is very limited real anomalous data.

The dataset used in this study only includes data on flight crashes, which is abnormal by nature. Therefore, synthetic normal data had to be generated to test the anomaly detection effectively. When combined with the real crash data, this synthetic data aids in the creation of a dataset that makes anomalies easy to spot and assess. Despite the original dataset has only anomalous crash data, this approach makes it possible to generate and validate anomaly detection algorithms within a controlled environment.

3.7.2 Train and Test Data Split

It's a technique for validating the model in machine learning where one divides the dataset into two subsets,

- Training dataset (80 %)
- Testing dataset (20%)

After fitting the model with a training set to allowing it to learn patterns from the data, and uses the test set to establish the model performance and its generalization capabilities. This will ensure that models make accurate predictions on new unseen data.

3.7.3 Fine-tuning and Hyper-parameter

The datasets may follow different distributions; the network of other tasks may be over-complicated for our tasks; or their structure may be too heavy. Therefore, we should fix the general layers of the pre-trained network and then modify the output layer to fit our task. (Wang & Chen, 2022)

Fine-tuning is a process where model parameters are slightly adjusted to optimize performance in a specific task. It will adjust the model to deal with some of the subtleties that the new data holds and that were otherwise ignored, hence generally enhancing its accuracy and performance for certain applications.

3.7.4 Predictive modelling algorithms

In our research on predicting flight crash risk, we will use different algorithms for machine learning to build predictive models that analyse such data on the visibility of the scene, crew age, and weather conditions to forecast a crash. All algorithms applied are explained in the following sections: K-Nearest Neighbors, Support Vector Machine, Logistic Regression, and Naive Bayes.

K-Nearest Neighbors (KNN)

K-Nearest Neighbors is one of the simplest yet very effective algorithms for classification and regression. For a prediction, KNN takes the majority vote of the k-nearest data points in the feature space. Since it is non-parametric, KNN does not require any assumption of the underlying distribution of data. The major reason why KNN is applied in anomaly detection in flight data is because it allows the identification of patterns by proximity. However, KNN can be very computationally heavy for large datasets and may perform poorly in high-dimensional data.

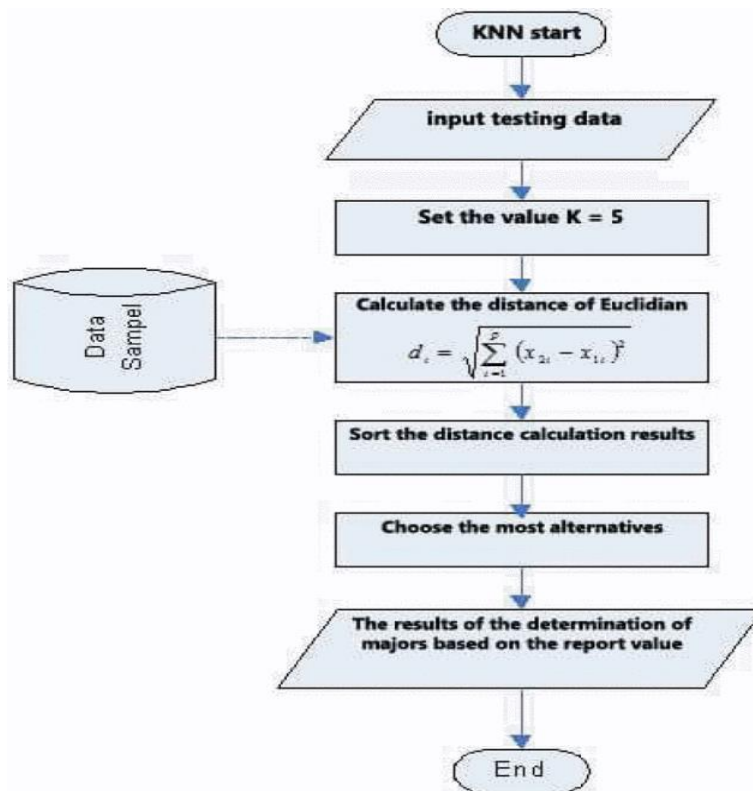


Figure 5 Flow chart explaining working of KNN (Akhiladevi M et al., 2022)

Support Vector Machine (SVM)

SVM is a very versatile algorithm that can handle both linear and nonlinear data by transforming the input region into a feature space with more dimensions. This is done using a kernel function, which maps the data to a higher-dimensional space where it is easier to separate the classes. (Ganesh Nimmada and S. Mahaboob Basha (2024)). In our research, SVM helps delineate the boundary between normal flight operations and potential crash scenarios. Its effectiveness is improved by working with kernels, which involves the transformation of data into higher dimensions to make the classification of nonlinear patterns easier.

Logistic Regression

Logistic Regression is a technique in statistics used to classify binary variables. This is modelling the probability of having a binary outcome dependent upon one or more predictor variables. This uses the logistic function to ensure the output produced is between 0 and 1. In the case of flight crash prediction, logistic regression helps in estimating the possibility of a crash based on features of the flights. It has the advantage that it is simple, interpretable, and comparatively efficient to train. On the other hand, logistic regression assumes a linear relationship between the input features and output log-odds, which might not be realistic in complex data sets.

Naïve Bayes

Naive Bayes is a simple probabilistic classifier based on Bayes' theorem and assuming independence between features. Shockingly simple, it often does really well in real-world problems, especially in large datasets. In this paper, the Naive Bayes algorithm will be used to predict the risk of a flight crash by calculating the posterior probability of crash given the flight data features. It has the benefits of fast computation with the ability to do binary and multiclass classification. However, the independence assumption made by it may sometimes prove unrealistic, hitting at the performance.

3.8 Justification for the Chosen Research Method

These different machine learning algorithms, including K-Nearest Neighbors, Support Vector Machine, Logistic Regression, Naive Bayes, among others, provide a rich methodological section for our research into the flight crash risk prediction effort. Such is the simplicity and power of KNN in anomaly detection, the resistance of SVM to overfitting and handling high-dimensional data, the interpretability and probabilistic framework of Logistic Regression, and computational efficiency of Naive Bayes for thorough identification of potential crash scenarios.

These algorithms will be used because of their individual strengths, which fit our diverse and complex dataset in a way that allows capturing linearity and non-linearity of the data. This synergy ensures a rigorous, transparent analysis with very promising results in our objective of predicting the risk of flight crashes by occurrence code and associated features. Among the many machine learning algorithms used, these methods yielded the most promising results, thereby showing their effectiveness in our research context.

Machine learning helps in extracting the relationships between the various factors either affecting or non-affecting the crash to the general information of the airplane and as a result, patterns are formed. Many researchers, in recent times, have been using several machine learning techniques to help the aviation industry and the professionals in determining the hurdles. Supervised machine learning algorithms like SVM, K-NN, ADABOOST and XGBOOST are used for the purpose of prediction. The work has helped in improving the accuracy to a great extent. (J. Raikar et al., 2020)

3.9 Alternative Research Methods for Predicting Flight Crash Risk

Advanced deep-learning techniques, such as convolutional neural networks and recurrent neural networks, are very proficient at capturing complicated patterns in high-dimensional data. Most importantly, among these methods is the ability to analyse a large amount of flight data and recognize advanced patterns indicative of possible crash risks. Their ability to auto-learn feature representations from raw data can get underlying patterns that traditional methods might overlook. Similarly, hybrid models which incorporate stacking and ensemble methods combine several algorithms to take advantage of their combined advantages, which lowers overfitting and enhances generalisation. These models improve predictive power and robustness by capturing various facets of the data.

Bayesian methods include Bayesian Networks and Bayesian Hierarchical Models, which allow for prior knowledge and uncertainty to enter the modelling process. These provide a probabilistic approach for making the predictions. This is critical in safety-critical applications like flight crash risk prediction, where handling of uncertainty and incorporation of domain knowledge is very essential. Time series analysis methods explicitly model temporal dependencies, which is very important in understanding the sequence of events that precedes a crash methods that include ARIMA and Seasonal Decomposition of Time Series. Finally, survival analysis techniques, such as the Cox Proportional Hazards Model and Kaplan-Meier Estimator, focus on the timing of events, providing insights into the risk factors and timing associated with flight crashes, allowing for more proactive risk management.

This paper thus concludes that deep learning approaches, hybrid models, Bayesian methods, time series analyses, and survival analyses are alternative methods generally providing robust frameworks for predicting flight crash risks. These will have varying strengths answering different facets of the problem, thereby enhancing overall reliability and accuracy in predictions. Combining these techniques could enhance the predictive models' performance even further and offer more profound insights into the elements influencing flight safety, guaranteeing a thorough examination and more successful risk management tactics in the aviation industry.

4. Analysis and Findings

4.1 Data Pre-processing & Exploratory Data Analysis

4.1.2 Detection and Correction of duplicate values

In the preparation of a dataset for research, we did column aggregation from different datasets and combined them to create one comprehensive dataset. The result was a dataset three times the size of the original with duplicate entries. For instance, each occurrence code, which uniquely identifies an incident, comes with information on the aircraft, crew, and weather involved. Given that each accident can be linked to multiple occurrence codes (where one code may lead to another) this merging process led to the repetition of the same incident multiple times, meaning many of the same incidents appeared more than once with different occurrence codes.

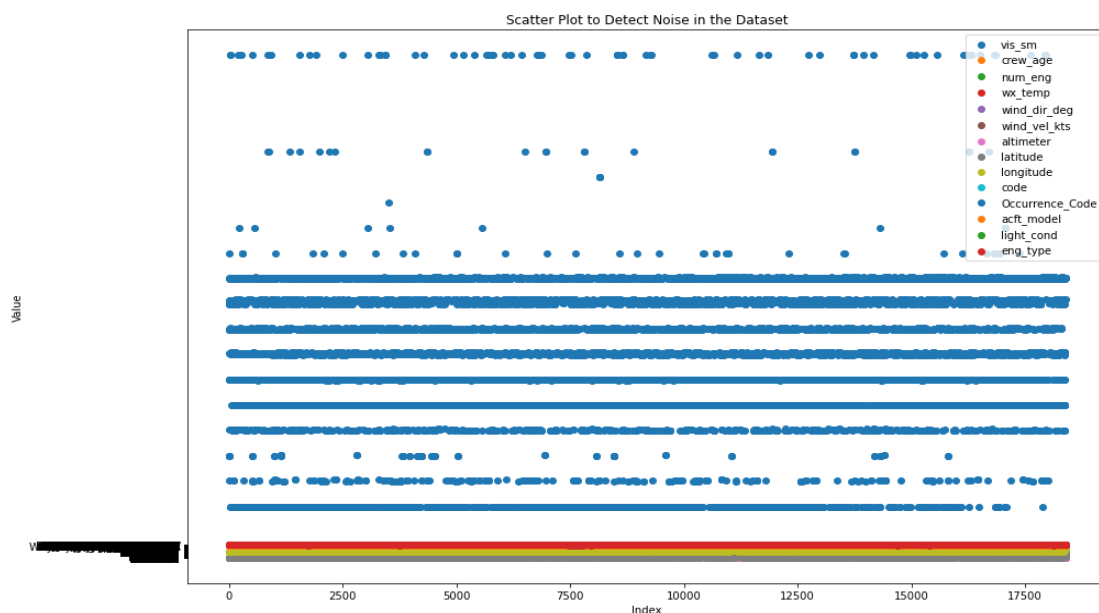


Figure 6 Scatter plot to detect noise

The plot includes a number of well-defined patterns and clusters; therefore, most of the points always remain aligned, which could be a case of redundancy. However, some isolated points and scattered values among the clusters indicate that there are variations or repeated entries. Variables like wind direction, temperature, weather conditions along with other variables show clear consistency and therefore duplicated values. It highlights how important preprocessing is in being able to deal with repeated values, providing results that are robust and reliable for later analysis.

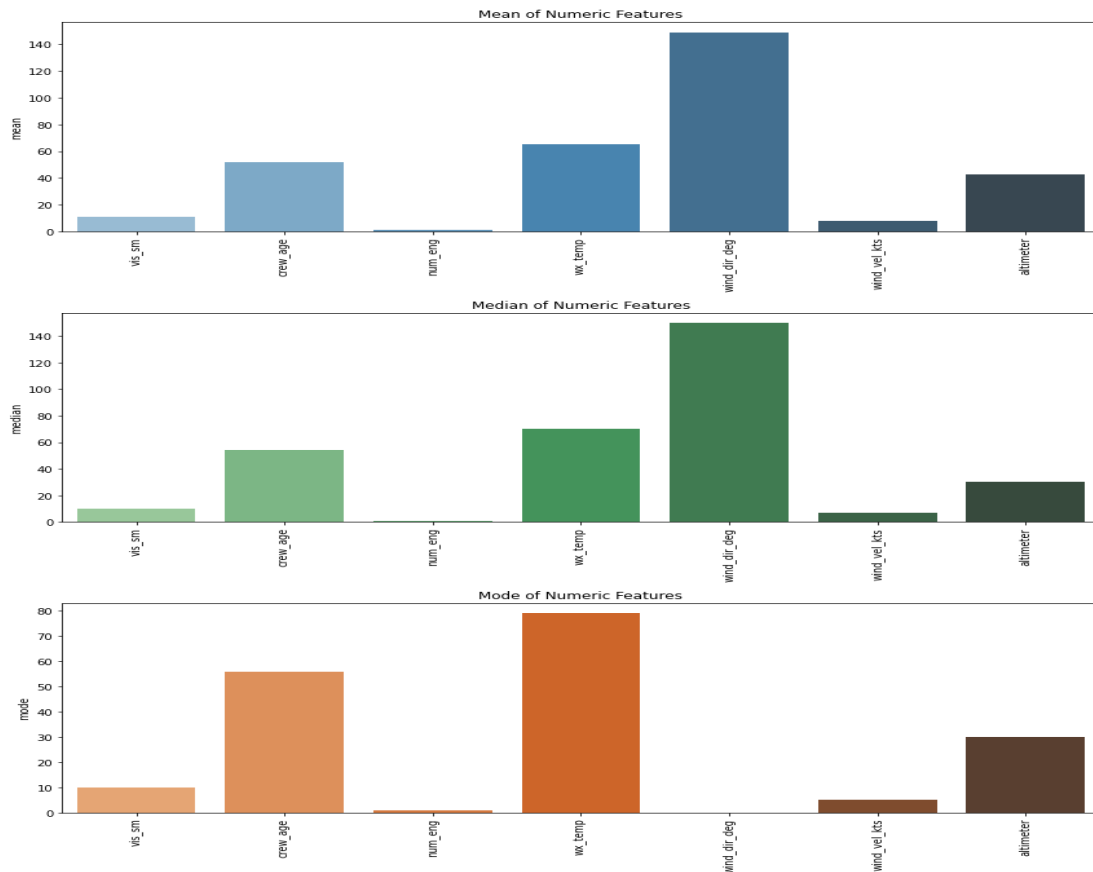


Figure 7 Computing statistical measures in numerical values

The plot allows the following inferences to be drawn:

- High mode values: Some features, like "vis_sm" and "crew_age", have very high mode values compared to their mean and median. This definitely indicates a high frequency of duplicate values for those columns.
- Closely laid out mean, median, and mode: Variables like "num_eng" and "wx_temp" have their mean, median, and mode quite close to each other in value. That would mean values are more spread, with less duplication.
- Mean Impacted by Outliers: Several times, the mean is way higher than the median, indicating that outliers are pulling the mean up. This can be seen with features like "wind_dir_deg" and "altimeter."

It provides an effective plot of the distribution of numeric features by comparing their mean, median, and mode. All these differences between these measures can give us insights into the duplicate values or outliers that might be present within a dataset.

The columns were merged on common identifiers such as 'ev_id' (Event ID) and 'Aircraft_Key' present in each dataset. We have applied the following adjustments to reduce duplicate values:

a) **Aircraft Category:** Originally, there were different kinds of aeronautical transports including helicopters, air balloons and etc. stored in the dataset. We filtered the dataset to only retain aeroplanes where 'acft_category' == 'AIR'.

b) **Occurrence Code:** Each occurrence is ordered in the column 'Occurrence_no'. We filtered only the first occurrence ('Occurrence_no' == 1), since the following occurrences are just consecutive steps of the same incident and, therefore, reflect a multiplication of accident data records.

c) **Engine Number:** The 'eng_num' column contains the number of engines with their details. In multiple-engine aircraft, all being of the same model, we considered only the record pertaining to engine number one; otherwise, we would have duplicate rows for the same model of the engine but different engine numbers.

d) **Aircraft_Key:** In case of accident, which included more than one collision of aircraft, different keys were given to these aircraft. The environmental data in such cases would be the same, but mechanical and crew data are different. So all data were kept.

After filtering using the above filters, dropping columns that were not needed was done to make the dataset leaner.

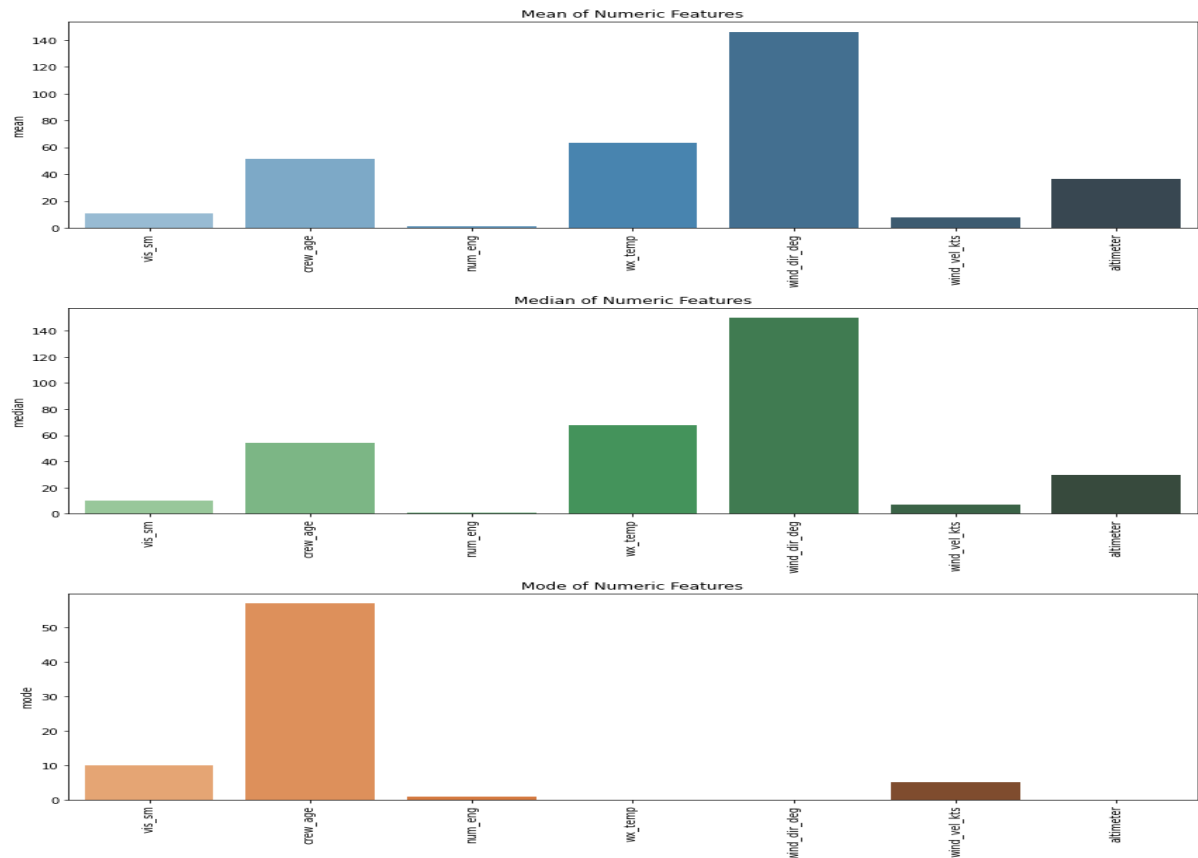


Figure 8 Computing statistics for each column after reducing the repeated values

1. Mode: The maximum change is noted in mode values. For most features, the mode has come down drastically. It means that the frequency of the most frequent value has decreased significantly, after removing the duplicate values. This explicitly explains that the dataset earlier contained a lot of repeated values.

2. Mean and Median: Compared to mode, the variations in mean and median are not that pre-eminent. This again makes sense, as the mean and median are not as strongly connected with the frequency of individual values as the mode is. Nevertheless, we still witness minor drops in these values, which convey to us that removal of duplicates has, at a minor level, affected the central tendency within the data.

4.1.2 Handling missing values

Given that the dataset used in this research is extensive and contains missing values amounting to 50% of the rows, ensuring completeness by addressing these missing values is a time-consuming and resource-intensive task, yet crucial for maintaining data accuracy and quality.

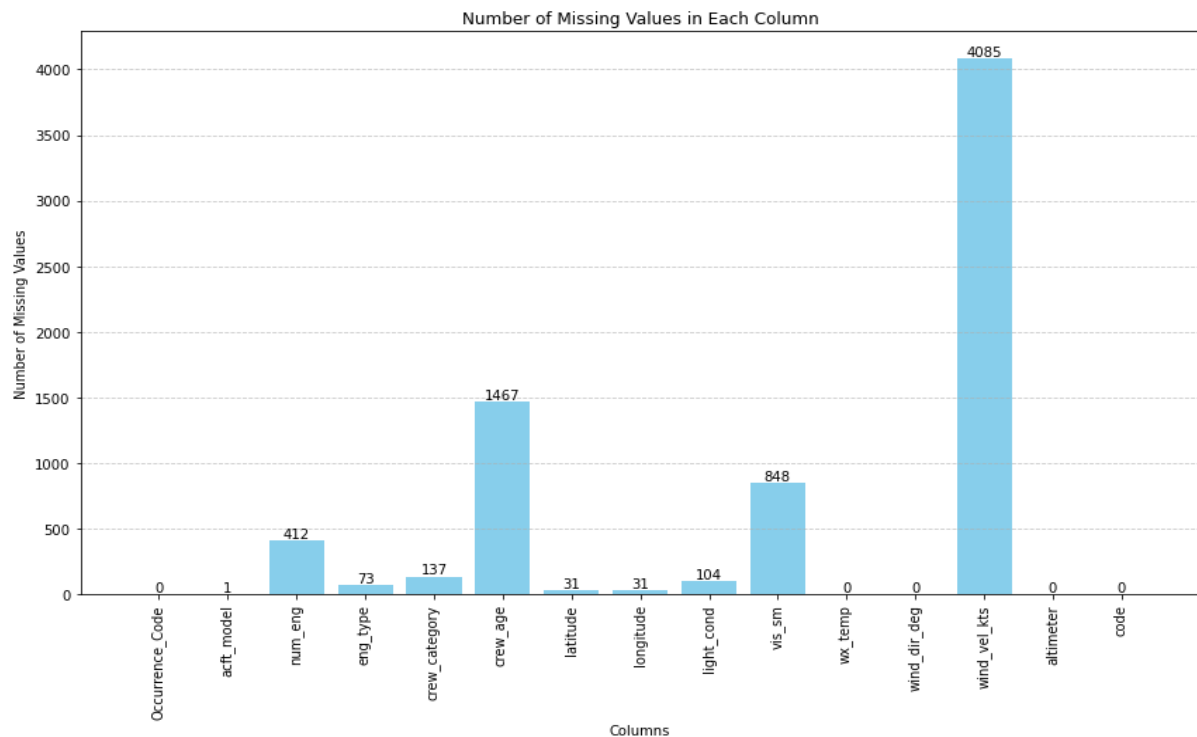


Figure 9 Number of missing values in each column

There are few missing values in the 'acft_model', 'latitude', and 'longitude' and it is difficult to fill them with meaningful values, those missing values have been dropped.

4.1.2.1 Filling the missing Values

a) num_eng and eng_type

The missing values in these features has been filled by having a dictionary that maps each unique aircraft model to its corresponding engine type and number of engines. First, the function identifies all the unique models and their corresponding engine details from the dataset. Then, a function is applied that checks each row for missing 'eng_type' or 'num_eng'. If not available, it checks in the model-to-engine information dictionary for these missing values and fills them accordingly. This process shall ensure that missing details on engines will be correctly filled based on known data for each aircraft model.

b) crew_category and crew_age

Decision tree classifier imputation exploits the predictive power and handling of complex relationships between features. Since it is trained based on existing values, the model learns patterns and predicts missing values according to some relevant features such as 'acft_model', 'eng_type', and 'crew_category'. In this regard, non-linear interactions and relationships that are normally difficult to capture with simple imputation methods are fully captured. In contrast to the naive use of the mean or median, decision trees are capable of utilizing several features for contextually accurate imputations, hence making this quite a robust choice for real-world datasets, which most definitely will contain varied data types.

c) vis_sm

The K-Nearest Neighbours (KNN) imputation approach is used to fill in the missing values in this column. It makes predictions about missing values based on how similar data points are. The technique uses features like 'wx_temp', 'latitude', and 'longitude' to approximate the values for the missing values in the column 'vis_sm'. To make imputation easier, categorical variables are first converted into a numeric format. Next, the dataset is subjected to an application of the KNN imputer with five neighbours, which uses the similarity between data points to fill in the missing values for 'vis_sm'. The categorical variables in the dataset are recoded to their original labels following imputation. By taking advantage of the links between the features, our method guarantees that the missing values are filled with estimates that are contextually relevant.

d) wind_vel_kts

The procedure entails impute missing values for features in this column using a RandomForestRegressor. To indicate missing data, blanks and 0 values in 'wind_dir_deg', 'wind_vel_kts', and 'altimeter' are first replaced with NaN. Using the available complete data, the RandomForestRegressor ensemble learning technique is used to anticipate these missing values. Using associations with other features, this method estimates the missing values in the target columns by training the model on rows with non-missing values.

Because RandomForestRegressor can simulate intricate, non-linear interactions between features, it is especially useful. This method is often robust and efficient for managing missing data. It also frequently yields accurate imputations.

e) Dropping remaining null values:

Even after algorithms fill missing values, some NaN values can remain due to incomplete data coverage, where predictors of imputation are missing; limitations in the models of imputation by themselves, which may not make perfect value predictions, especially in complex scenarios, or data integrity issues like entry errors. Also, edge cases or outliers might not be well-represented, and algorithm constraints can leave some values unfilled. Those values are less in number and they are dropped from the dataset.

4.1.3 Outliers detection and removal

An outlier is an observation that lies an abnormal distance from other values in a random sample from a population. (Engineering Statistics Handbook, 2019)

The Z-score is a way to figure out how far away a piece of data is from the average of a group, measured in standard deviations. It tells us if a data point is typical or unusual compared to the rest of the group, which is useful for spotting unusual values and comparing data between different groups. (*Z-Score: Meaning and Formula*, n.d.). The Z-score is a way to find out how many standard deviations a data point lies from the

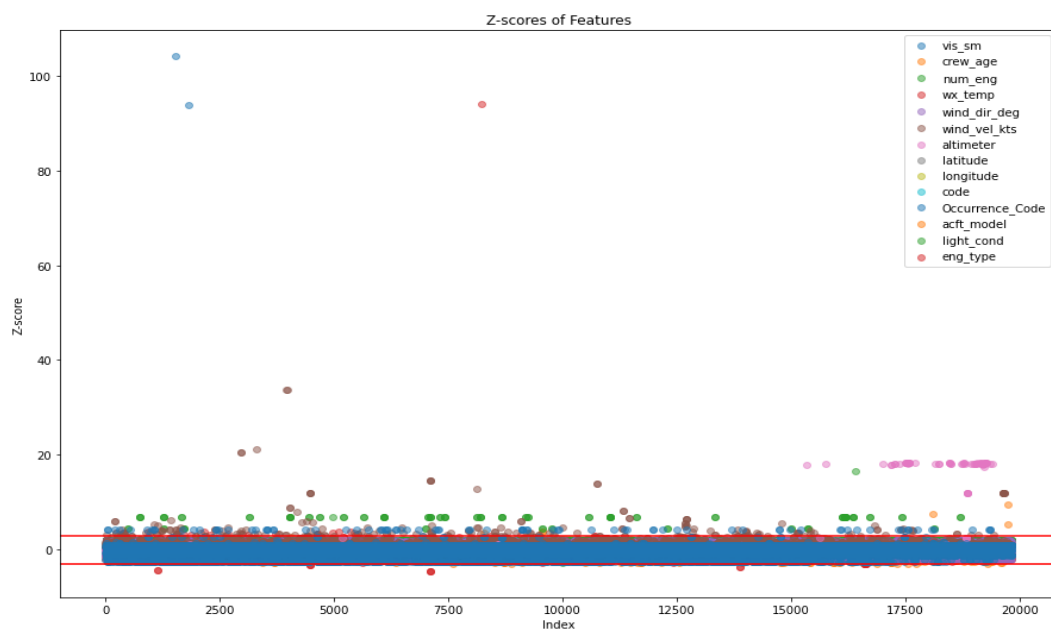


Figure 10 Z-score for each column

Multiple data points over the outlier threshold are displayed in 'vis_sm', 'crew_age', 'num_eng', and 'wx_temp', indicating the prevalence of outliers in these characteristics. These outliers have different distributions; for example, 'wind_vel_kts' features may have a few extreme outliers, whereas 'crew_age' features show a cluster of outliers within a particular range. The substantial prevalence of outliers points to possible problems with the quality of the data, which could be caused by inaccurate measurements, mistakes in data gathering, or real extreme values. It is vital to handle these outliers in order to guarantee the prediction model's accuracy and dependability.

Plots depicting outliers in each column:

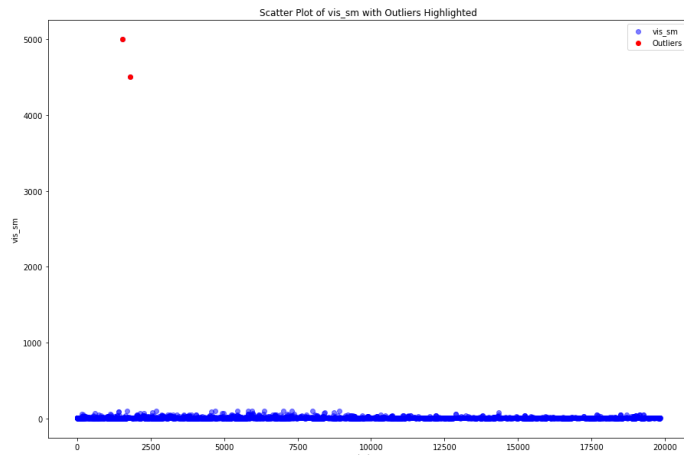


Figure 11 i) Outliers in vis_sm

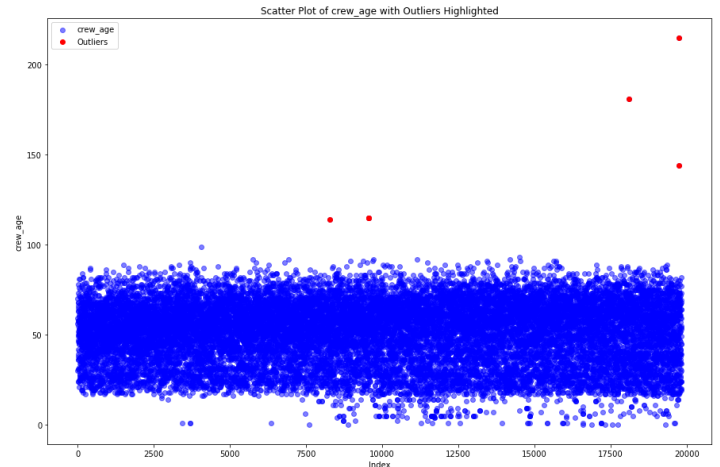


Figure 11 ii) Outliers in crew_age

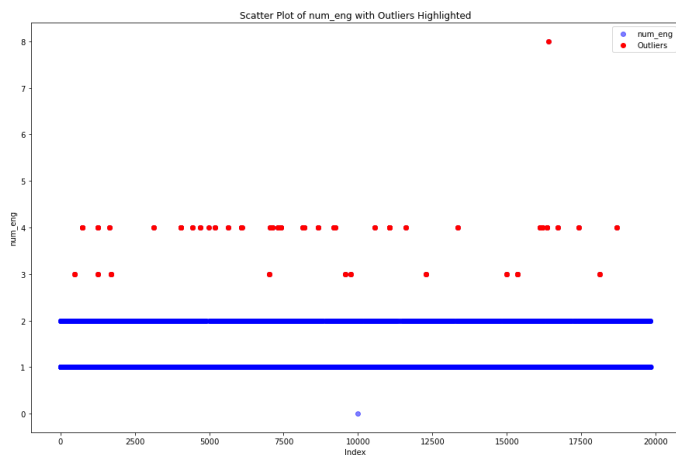


Figure 11 iii) Outliers in num_eng

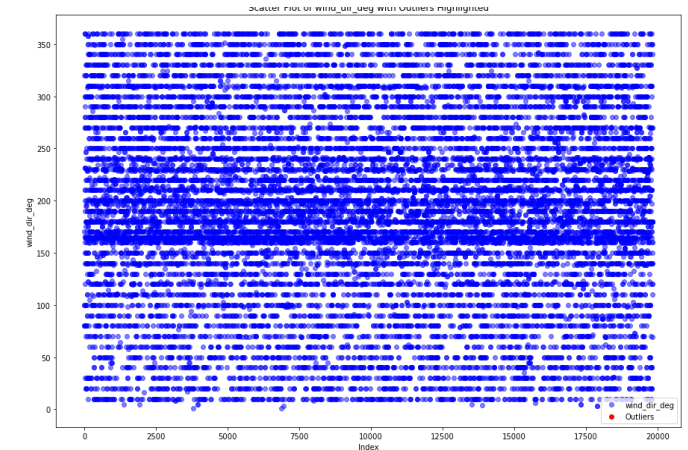


Figure 11 iv) Outliers in wind_dir_deg

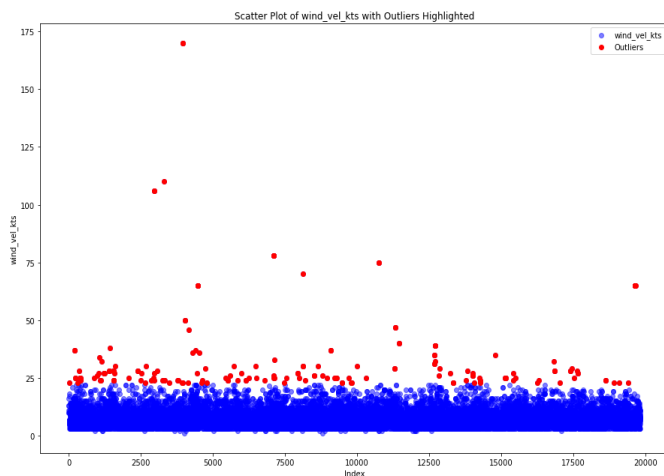


Figure 11 v) Outliers in wind_vel_kts

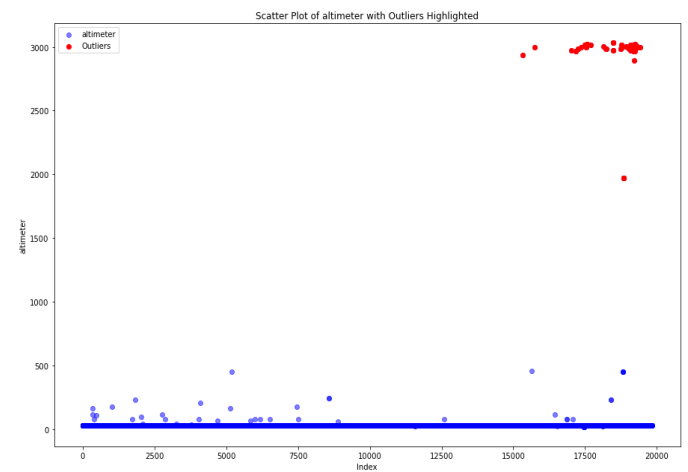


Figure 11 vi) Outliers in altimeter

There were very few number of outliers present in the dataset in each column. By identifying them using Excel, Python was then used to eliminate these, creating a clean dataset suitable for further investigation.

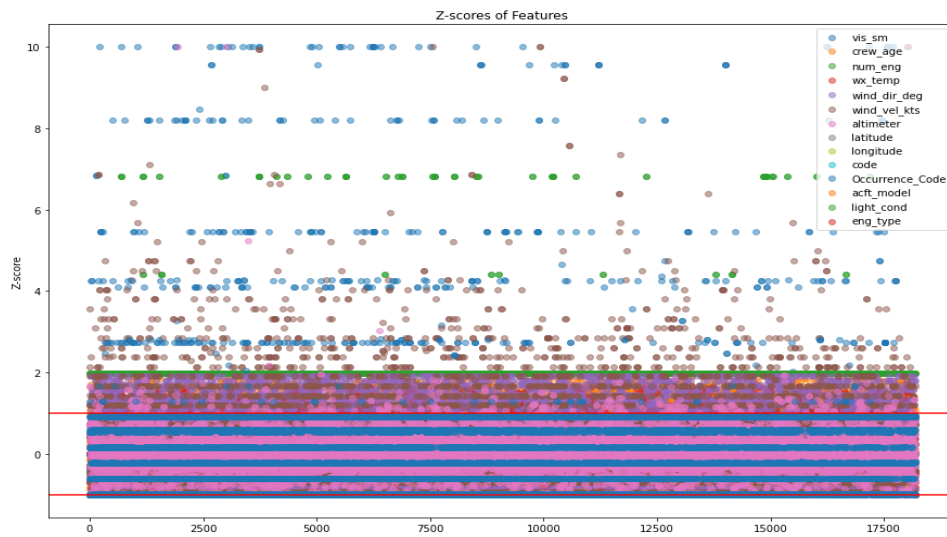


Figure 12 Z-Score after removing the outliers

4.1.3 Insights of the Dataset

a) Wind speed

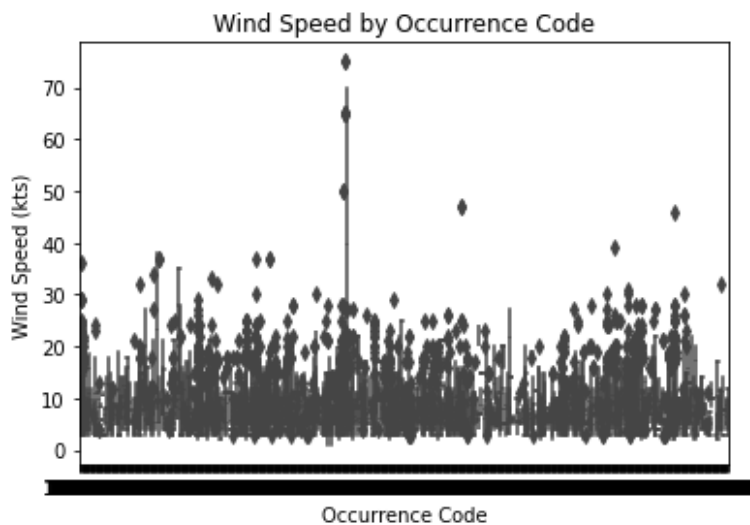


Figure 13 Wind speed by Occurrence

This distribution is very right-skewed, meaning most wind speed readings are less than 50 knots. The median wind speed appears to be around 10 knots. There seems to be a lot of variability in the wind speed, some occurrence codes demonstrate this variability more than others do. The shape of the distribution of wind speeds may indicate prevailing weather in the region. Preliminary look at the outliers suggests that data quality is quite perfect.

b) Visibility based on the light condition

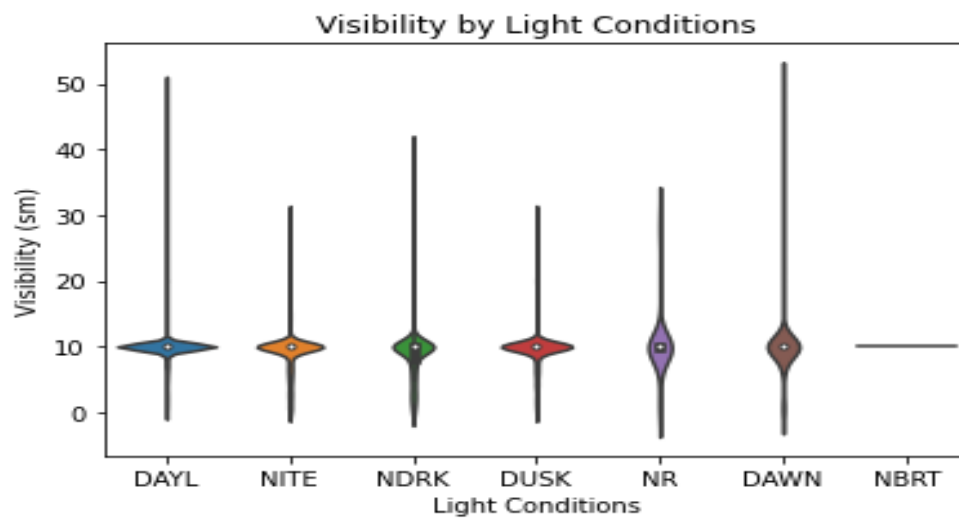


Figure 14 Visibility by light conditions

In most aspects, the visibility under daylight conditions is generally higher than under other light conditions. Relative visibility under conditions of night, NITE (night), DUSK and DAWN, is lower than in daylight. For most light conditions, the distribution of visibility is skewed to the right, indicating that there are more observations of lower visibility. There are outliers within some of the light conditions, particularly NITE and DAWN, indicating that there were instances of very poor visibility in those times.

c) Time series plot

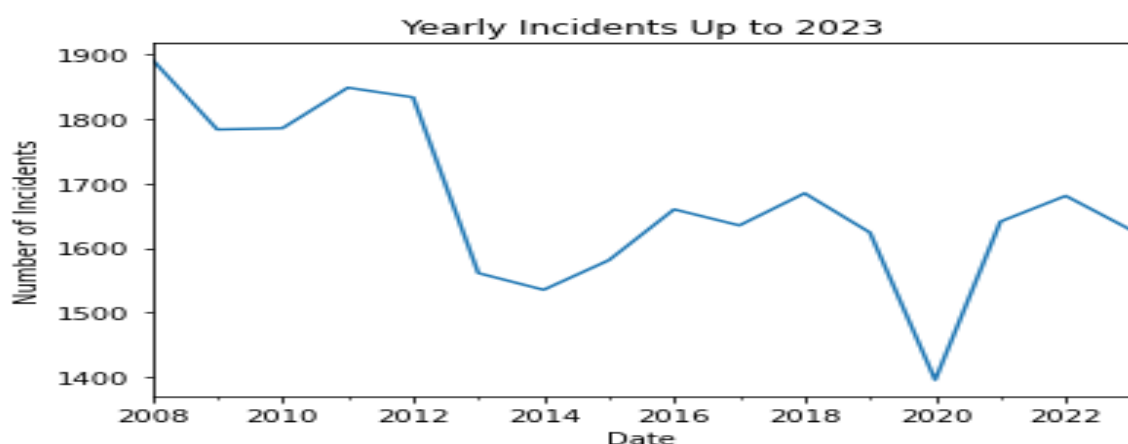


Figure 15 Yearly incident up to 2023

The plot depicts the number of incidents yearly which gradually decreases year by year, ensures the improvements in the safety measures. The gradual fall between 2019 and 2020 indicates the COVID period when the minimal air transportation taken place.

d) Flight Crew age distribution

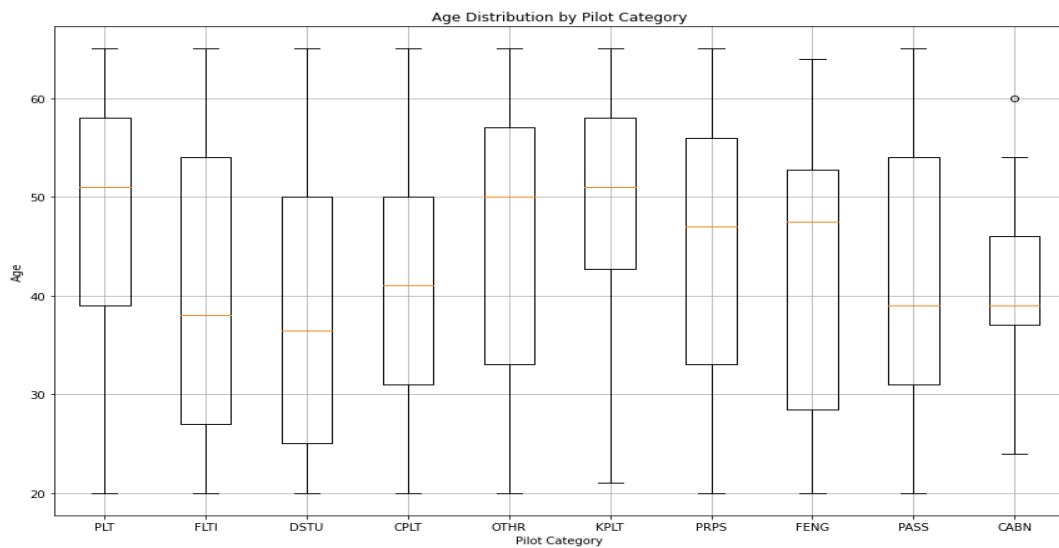


Figure 16 Age distribution for each crew category

The boxplots show the dispersion of the crew's ages in relation to the various pilot categories. In the vertical axis, age is represented, and in the horizontal axis, the pilot categories are. The line inside each box gives the median age for a certain pilot category. There appears to be a median age varying across the categories, with some of them above the general median age of others. Outliers are the individual data points plotted beyond the whiskers in a plot, and in general they are not particularly evident in this plot. These would be the outliers, which are crew members whose ages are either very high or low compared to others within the same category.

Table 3 Crew category code explanation

Crew Category	Code
Co Pilot	'CPLT',
Student Pilot	'DSTU',
Flight Engineer	'FENG',
Flight Instructor	'FLTI',
Check Pilot	'KPLT',
Other Flight Crew	'OTHR',
Pilot	'PLT',
Captain	'CBTN'

e) Visualizing Anomaly Patterns with t-SNE

t-SNE is a nonlinear technique that focuses on preserving the pairwise similarities between data points in a lower-dimensional space. t-SNE is concerned with preserving small pairwise distances. t-SNE preserves the relationships between data points in a lower-dimensional space, making it quite a good algorithm for visualizing complex high-dimensional data. (Awan, 2023)

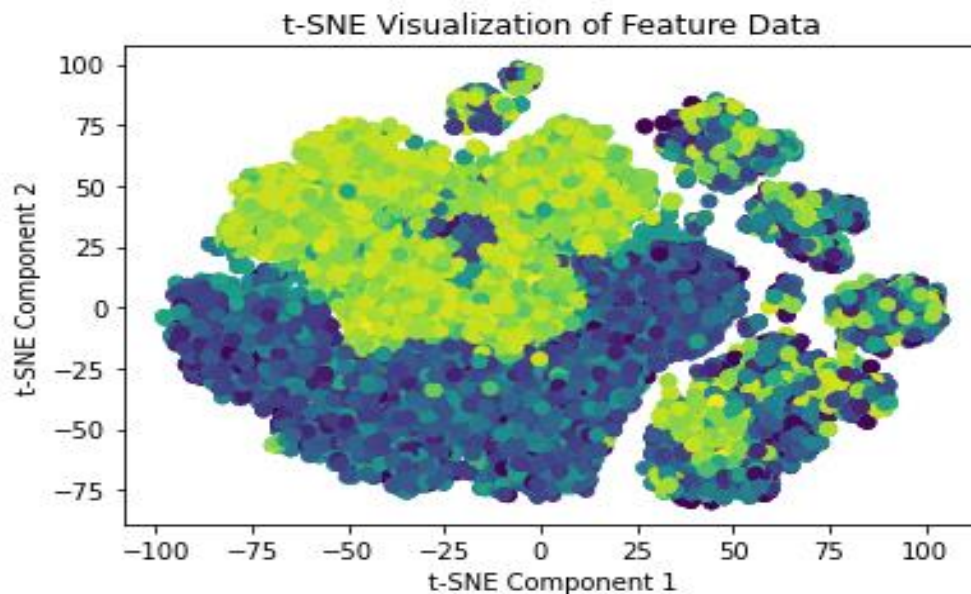


Figure 17 t-SNE visualization of Feature Data

This t-SNE visualization will reduce the high dimensionality of the anomaly data down to two dimensions so that possible relationships between different anomaly types, colored by the 'Occurrence_Code' feature can be seen. These points tend to form a number of distinct clusters, notably a very prominent blue cluster toward the bottom left and a large green cluster in the centre. It suggests that perhaps there is some deeper structure or groupings within the anomaly data. That is, it indicates that different occurrence codes are likely to be correlated to different patterns in the high-dimensional data. The data points do not scatter uniformly across the plot. There are denser areas with more concentrated data points in clusters, and sparser areas with fewer data points. It may show subgroups in these large clusters or more specific types of anomalies. Generally, there are not that many outliers, except for some points that seem isolated from other main clusters.

4.2 Multivariate Analysis

Most datasets involved in aviation pertain to variables dealing with environmental, aircraft specifications, and operational factors, all relating to each other in complex ways. Proper handling and interpretation of these large-scaled complex data are required in advancing statistical techniques and state-of-the-art machine learning methods. Multivariate analysis is thus a keystone to extracting meaningful patterns and relationships from this very data. In the process of analysing complex aviation data, a good number of statistical and machine learning techniques are applied to ensure that the dataset is manageable yet informative for predictive modelling. Specifically, steps taken to make the dataset refined for this include Principal Component Analysis, Discriminant Analysis, and clustering methods, which are explored in this paper. First, it was correlation analysis; then PCA with further refinement using discriminant techniques to choose the number of components.

4.2.1 Correlation Analysis

The preliminary step involved a thorough correlation analysis of the dataset. The correlation matrix was computed to understand the relationships between various variables. However, the correlation analysis revealed a generally poor correlation between the variables. This outcome can be attributed to the nature of aviation data, which is often vast and complex, encompassing a wide range of variables and contexts. Specifically, aviation datasets can be characterized by numerous features with intricate interactions, leading to weak linear correlations when considered in isolation.

The correlation matrix shows that nearly all features are weakly correlated. Their values are very close to zero, which might indicate that most features are independent of one another. A strong positive correlation between the number of engines and type of engine, other positive correlations such as wind direction and wind velocity, wind velocity and number engines, aircraft model and the crew age are relief.

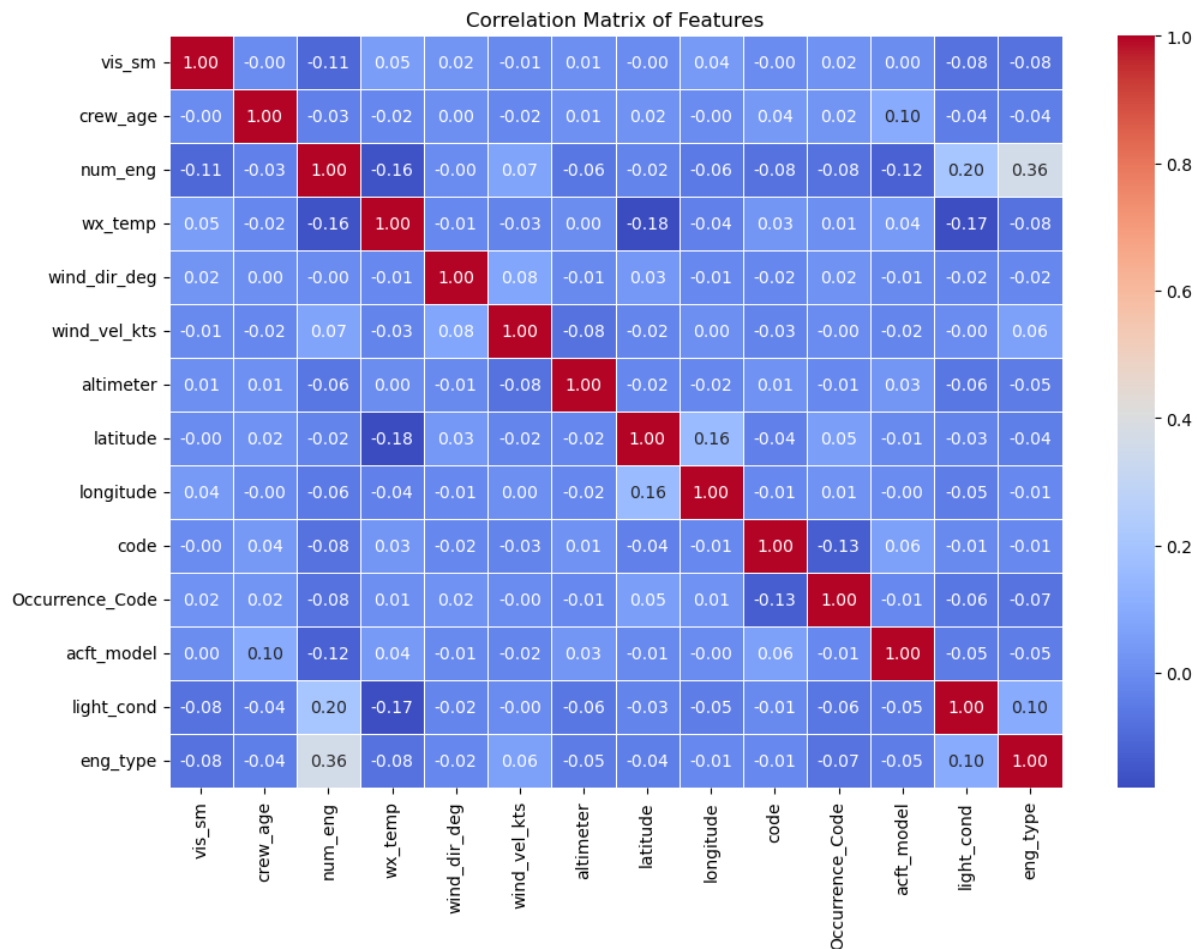


Figure 18 : Correlation Matrix of Features

These results show an indication that although there is some relative relationship with some features to some extent, no strong multicollinearity exists, which is good for statistical modelling. These low correlation values indicate that most of the features can be used in models with no significant multicollinearity. However, the lack of strong linear relationships may present challenges to predictive modelling and call for careful feature engineering. Here, PCA can prove important in dimensionality reduction and may further reveal patterns that were otherwise hidden. In transforming features into uncorrelated principal components, this would improve the model performance and its interpretability. The development in the future would turn to using PCA to further understanding and deal with the complex and high-dimensional dataset.

4.2.2 Principal Component Analysis (PCA)

PCA was applied to reduce the dimensionality of the dataset while retaining the most significant features that capture the maximum variance. The goal was to simplify the dataset and enhance interpretability by transforming the original variables into a new set of uncorrelated variables called principal components.

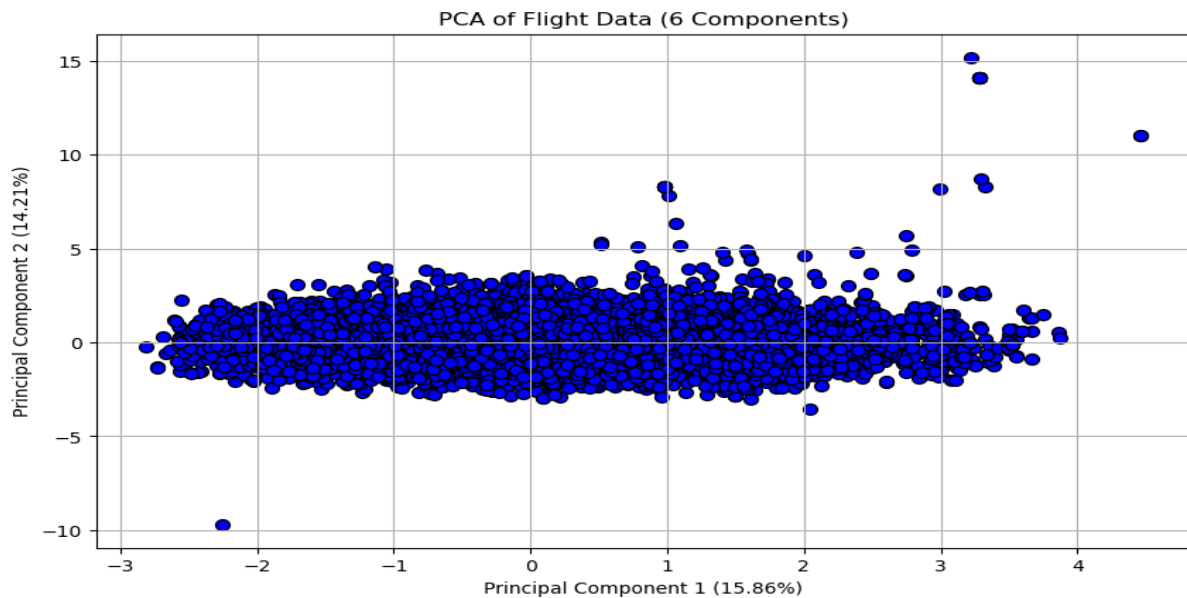


Figure 19 PCA of the data

Initial PCA results showed that the first two principal components cumulatively explained only 29% of the variance in the dataset. This level of variance was deemed insufficient for reliable validation and training of the model. In practical terms, capturing less than 30% of the variance suggests that critical information is still missing. It is observed that the PCA values are less due to the selective nature of variables chosen. This reduced set of features might not hold the variability to its full extent present in the dataset and hence lower explained variance by principal components.

PCA Cumulative Explained Variance:

```
Number of Components: 1, Cumulative Explained Variance: 0.1586
Number of Components: 2, Cumulative Explained Variance: 0.3008
Number of Components: 3, Cumulative Explained Variance: 0.4352
Number of Components: 4, Cumulative Explained Variance: 0.5601
Number of Components: 5, Cumulative Explained Variance: 0.6802
Number of Components: 6, Cumulative Explained Variance: 0.7935
Number of Components: 7, Cumulative Explained Variance: 0.9043
Number of Components: 8, Cumulative Explained Variance: 1.0000
```

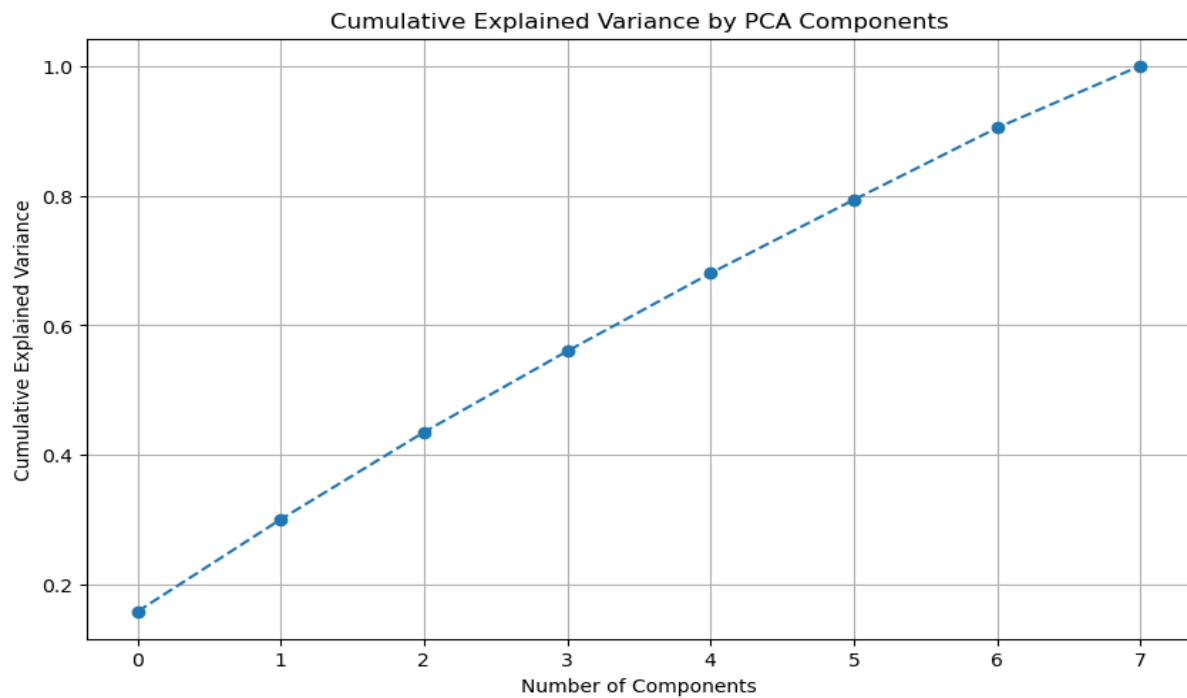


Figure 20 Cumulative explained variance by PCA components

The curve finally plateaus somewhere around 1.0, indicating that about 5-6 components already explain almost all variability in the dataset.

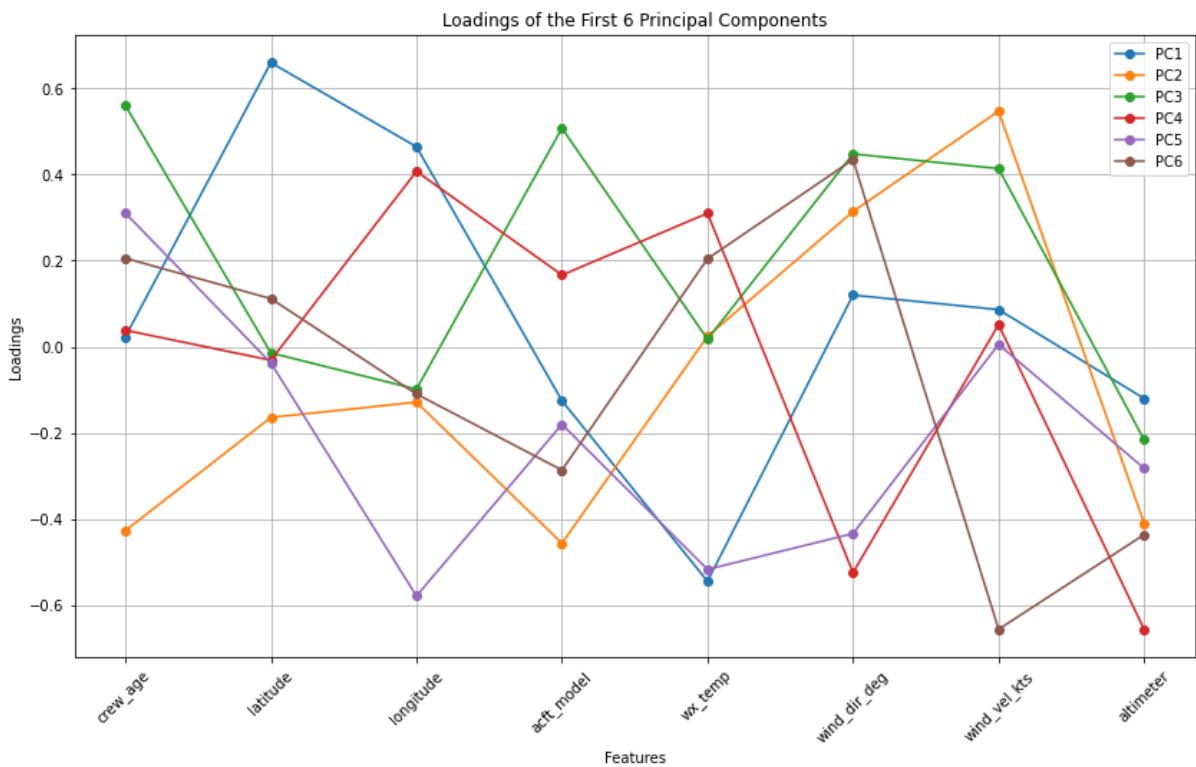


Figure 21 Loadings of PCA components

The contribution of each input feature to the first six principal components is shown in the PCA loadings plot (Figure 20). One can see that the crew's age has quite a strong positive loading of about 0.6, thus it would be one of the major factors within this component. Latitude has a positive loading of about 0.4 on PC1 and some substantial loading on the second principal component, about 0.6, so it is influential in both components. Longitude positively contributes with a loading of about 0.6 to PC3. The aircraft model has a positive loadings on PC2 and PC3 but negative on PC4, so it has at times variable influence across many components. 'wx-temp' has a negative loading of about -0.4 on PC2, while it varies in other components. The wind direction in degree has a positive contribution to PC3 and a negative contribution to PC5, hence manifesting its relation with multiple principal components. Last, the wind velocity in knots had positive loadings on PC2 and PC6, while altimeter contributed negatively to PC1 and PC4 but positively to PC5. These patterns underline what dominant features each principal component represents, thus indicating how important they are in explaining the total variance of a data set and proving useful for further analysis.

Such an observation underscores how efficient PCA is at reducing data complexity while keeping a high rate of information retention. It enables efficient dimensionality reduction, most often simplifying models and improving computational efficiency without the significant loss of data integrity. One of its positive aspects is that it is capable of capturing most of the variances with only a few components, hence enabling easy interpretation and analysis. The first few principal components will cover most of the variability due to the concentration of major patterns in the data. Often this is because of redundant or correlated features in the dataset, letting a few components catch most of the dominant variability. This is evidenced through the steep initial rise of a cumulative explained variance plot, followed by levelling off as more and more components are added.

4.2.3 Factor Analysis

Factor analysis has thus been applied in this research to reduce a large and complex set of data from the aviation domain down to some smaller number of underlying factors. From an examination of the factor loadings, one is able to see how various characteristics contribute to these factors, thereby helping gain an insight into the structure of the dataset.

Factor Loadings:

Table 4 Factor loadings

	Factor1	Factor2	Factor3	Factor4	Factor5
crew_age	0.119496	-0.030727	-0.012557	0.051161	0.008249
latitude	0.041989	-0.015396	0.535959	0.411381	0.065864
longitude	-0.011665	0.034041	0.266236	0.039496	-0.022094
acft_model	0.886976	-0.007689	0.046332	-0.169248	-0.019654
wx_temp	-0.010118	-0.015918	-0.092371	-0.318832	-0.009749
wind_dir_deg	0.008638	0.057392	-0.011912	0.020918	0.494343
wind_vel_kts	0.002377	0.714550	-0.082814	0.071507	0.068642
altimeter	0.033580	-0.114927	-0.049288	0.012426	-0.009022

The factor loadings denote the strength by which each variable contributes to the identified factors. For instance, `acft_model` has a high loading on Factor 1 of 0.887, which means it is very strongly related to this factor. On the other hand, `crew_age` indicated very minimal loading across all factors, indicating it had less influence in the factor structure. Variables like `acft_model` strongly relate to Factor 1, `wind_vel_kts` to Factor 2. Latitudes, longitudes, and wind directions in degrees are related to factors 3, 4, and 5, respectively, which uncover various dimensions of the aviation data.

Dimensionality reduction in a dataset using factor analysis ensures that relevant variability is retained. It helps in understanding the latent structure of data and allows key variables to be identified that drive the observed patterns. This simplification is critical to effective model building and interpretation.

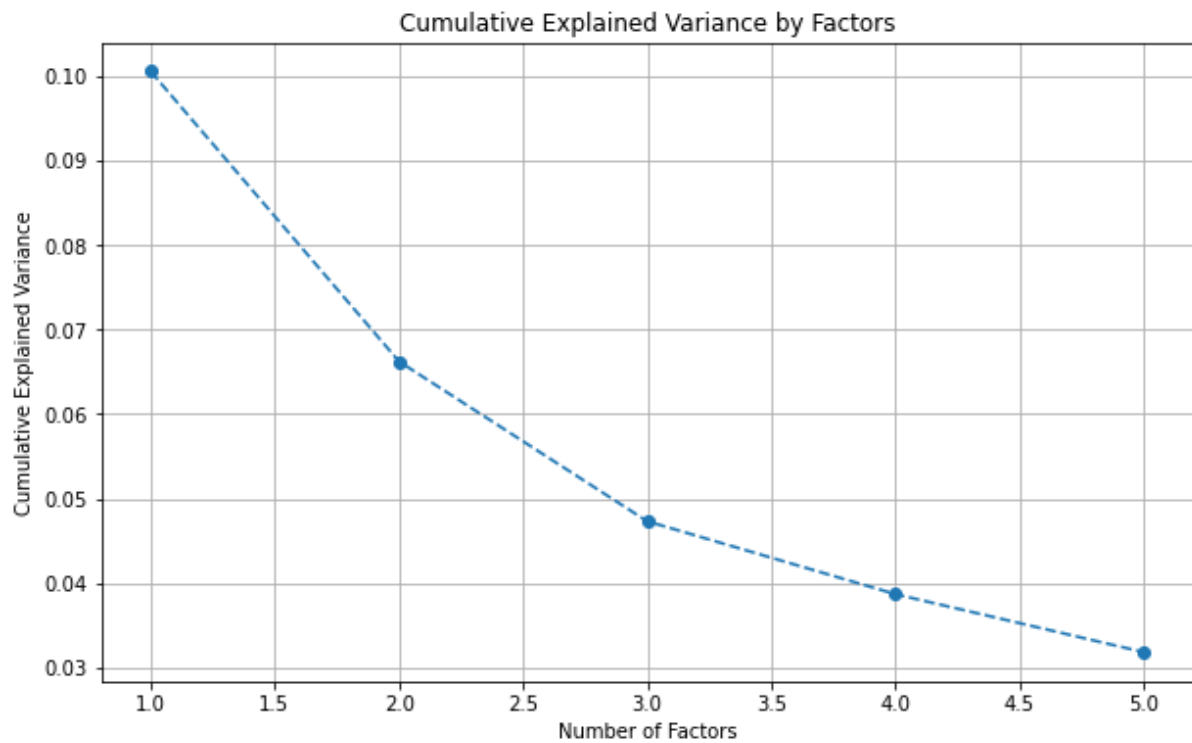


Figure 22 Cumulative Explained Variance by Factors

Factor analysis plots indicate that there will be a steep initial rise of the cumulative explained variance, hence the first few factors capture a significant portion of the variability in the data. This means that only a limited number of factors are both necessary and sufficient to summarize the data set, hence improving interpretability and simplifying the model. The plot levels off, eventually suggesting that beyond some number of factors, additional ones add very little insight.

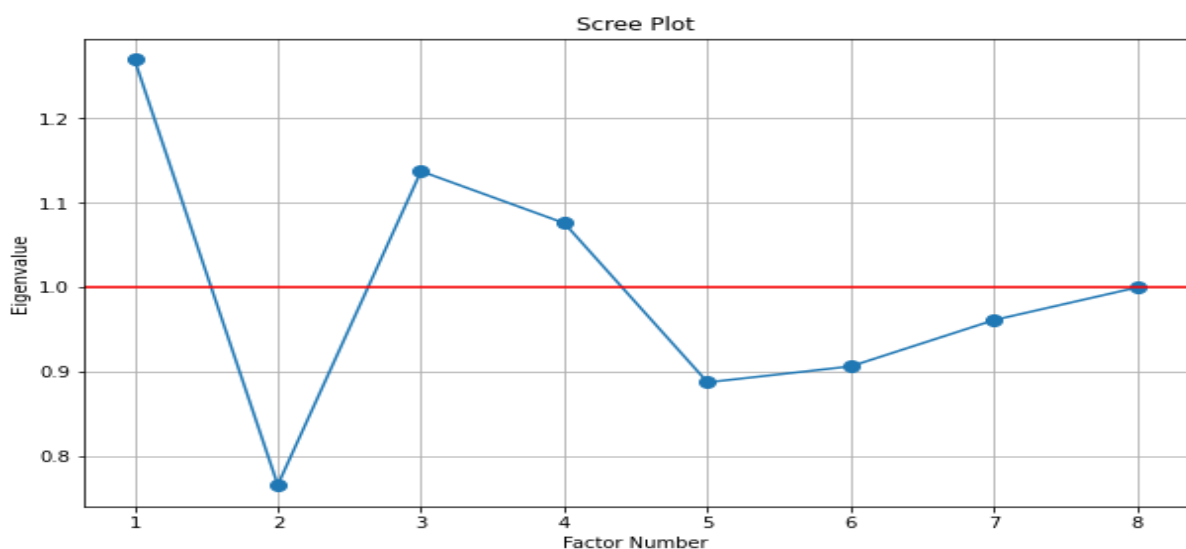


Figure 23 Eigen value for each Factor

The scree plot might insinuate important statistical patterns in eigenvalues derived from the correlation matrix. It shows remarkable insight into data structure. That steep initial drop in the eigenvalues means that just the first few of those factors capture a large fraction of total variance, hence establishing their importance. There is a visible "elbow" at which there could be a potential cut-off point after the second factor, where the rate decreases, and indicating diminishing returns from further factors. The red horizontal line at an eigenvalue of 1 serves as a threshold, identifying significant factors. Factor 1 is greater than 1.2 and thus captures a major portion of variance; Factor 2 drops below 0.8, but still conveys less but relevant information. Factors 3 through 8, with eigenvalues of approximately or a little above 1, contribute variably, hence marking them either as substantial or marginal. It is recommended to retain all the factors above the threshold since this would balance model complexity with interpretability, making further efficient and accurate analyses focusing on a few informative components.

Factor analysis, therefore, enhances interpretation and management capabilities of aviation accident data, in particular complex ones, as it shows hidden structures and relationships. Despite its strengths, it should be used in conjunction with other methods and domain knowledge to ensure comprehensive analysis.

4.2.4 Cluster Analysis

With the dimensionality reduced to six principal components, the next step involved preparing the dataset for clustering and further analysis. K-means clustering was applied to group similar instances within the dataset. Standardizing the data ensures that each feature contributes equally to the distance calculations used in the clustering algorithm. Euclidean single distance calculation method is approached here. This step helps in identifying distinct clusters or patterns within the data, which can provide valuable insights for further analysis.

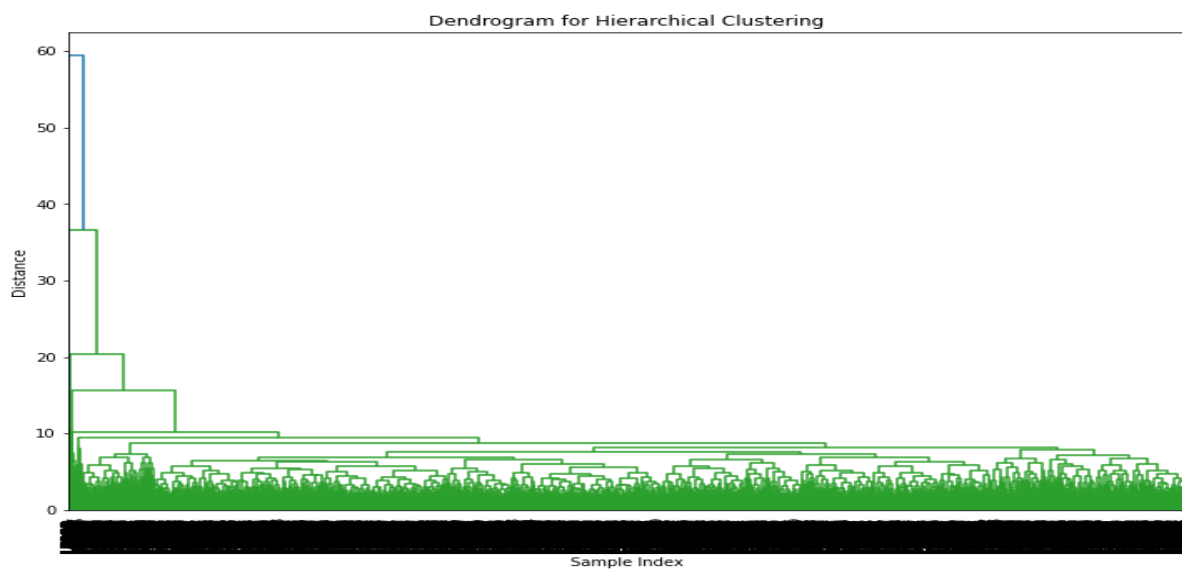


Figure 24 Dendrogram for Hierarchical clustering

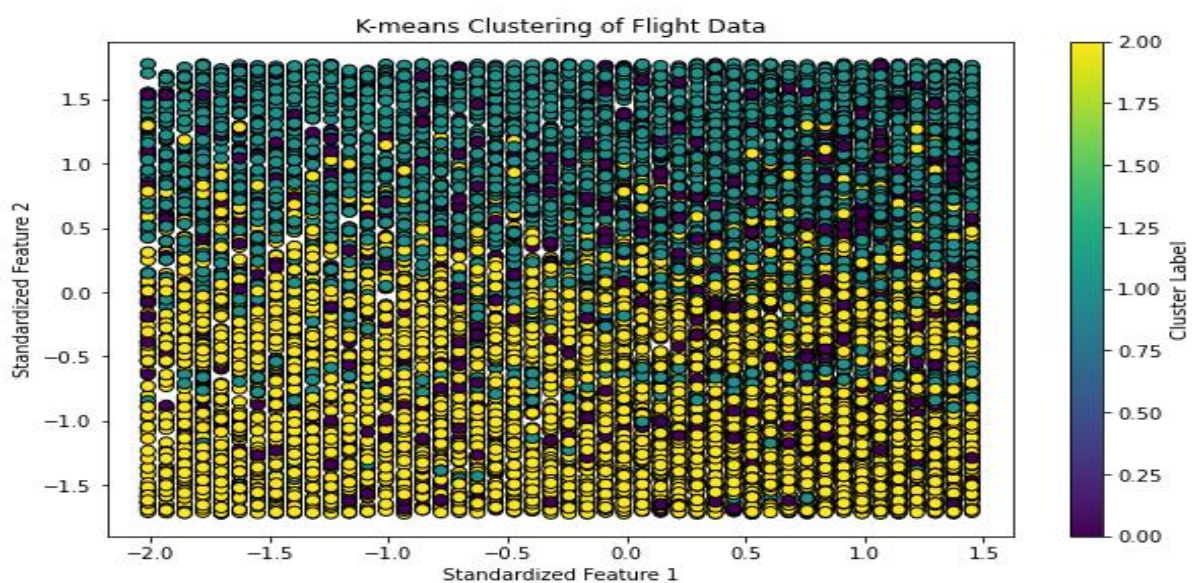


Figure 25 K-means Clustering

Overview of the Cluster Centroids

The cluster centroids plot, derived from the K-Means clustering algorithm with two clusters, has prominent differences in the characteristics of the two identified groups as shown:

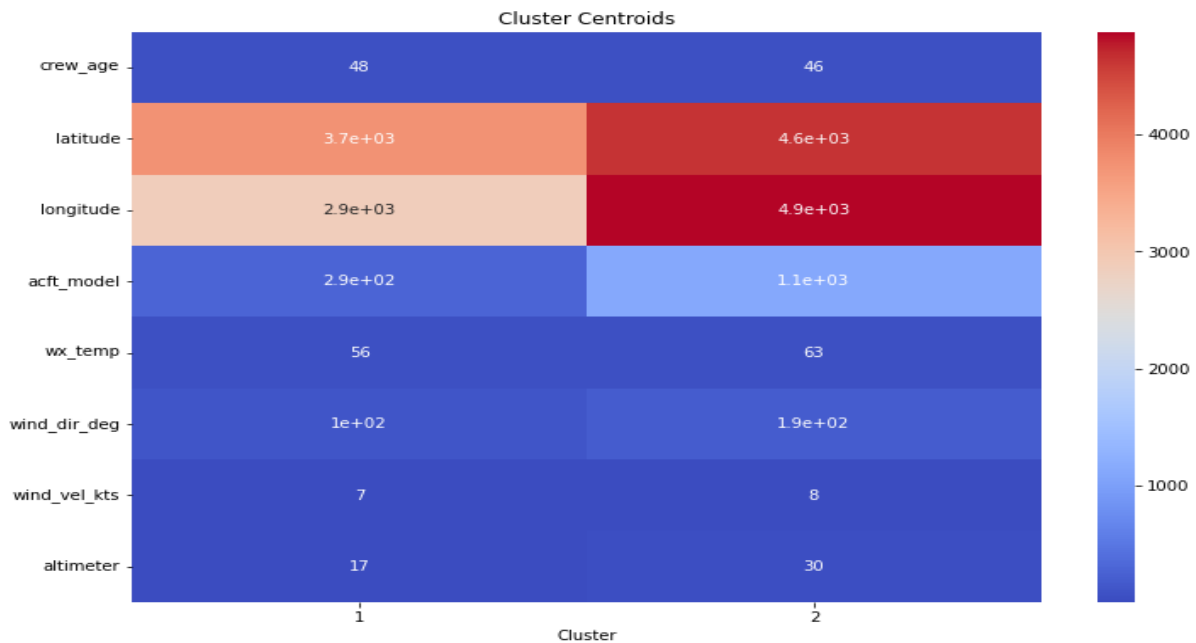


Figure 26 Cluster centroids

- Cluster 1: This cluster generally has lower values on 'crew_age', 'latitude', 'longitude', 'acft_model', 'wx_temp', 'wind_dir_deg', 'wind_vel_kts', and altimeter across most variables. This likely indicates that Cluster 1 is formed by flights involving small aircraft flying at lower latitudes in cooler weather with light winds and at lower altitudes.
- Cluster 2: This cluster has higher values for exactly the same variables, and thus it is going to include flights with larger aircraft, operating in higher latitudes, warmer weather conditions, and facing higher wind speeds with altitude.

These results might point to the fact that this clustering algorithm has made a good differentiation of flights by key operational and environmental factors. These clusters clearly correspond to different flight types, such as domestic versus international or short- versus long-haul, which have very distinct operational contexts again.

Silhouette Scores Analysis

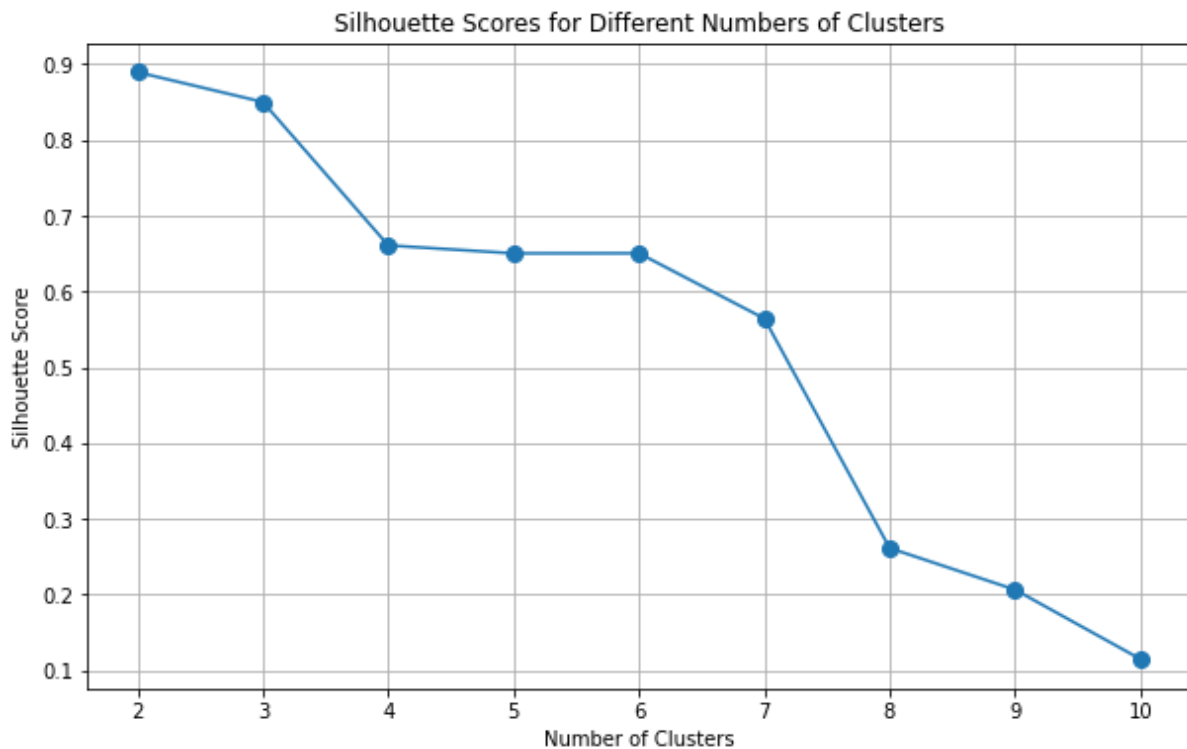


Figure 27 Silhouette scores for Different numbers of Clusters

The silhouette scores give an insight into the quality of the clustering outcome for different numbers of clusters:

- Two Clusters: Silhouette score is 0.849, which means very high separation and good cohesion. This high score suggests that the two-cluster solution is robust and does represent well-defined groups within the data.
- Three Clusters: The silhouette score increases very slightly to 0.850, showing there is even better separation and cohesion with an additional cluster. This minor improvement may indicate the third cluster allows for a gradient-like understanding of the data, but overall, the increase is marginal.
- Four or More Clusters: The silhouette score decreases as the number of clusters goes above three. For four clusters, this goes down to 0.716 and further declines as the number of clusters increased to 0.714, 0.642, 0.597, 0.522, and remained stable at 0.522 for ten clusters. These decreasing silhouette scores indicate that added complexity is not giving significantly better separation between clusters.

The clustering analysis, more specifically, the two-cluster solution, does efficiently identify distinct groupings of the crash data based on aircraft characteristics, geographic locations, and weather conditions. The high value of the silhouette score for the two-cluster solution supports its robustness and relevance. However, for a number of clusters higher than three, there is more complexity without substantial improvements in clustering quality. Therefore, the two-cluster and three-cluster models are most appropriate for further detailed analysis, with possible implications for the targeting of safety measures and the understanding of different crash scenarios.

4.2.5 Conclusion of the use of Multivariate analysis in predictive modelling

In our research related to aviation crashes, several statistical analyses have been applied with the twin purposes of improving predictive modelling and discovering hidden patterns in data.

First, correlation analysis was done to see the relationships between the features. The result was weak correlations and no strong multicollinearity. This is advantageous in model development since it shows most of the features can be used without significant redundancy. The fact that only weak linear relationships existed led to the use of advanced methods so that more complex interactions would be captured. For that, Principal Component Analysis was done to identify the structure of the dataset better.

It was instrumental in reducing the dimensionality using PCA, which showed that only a few of the principal components captured a large part of the variance in the data. More specifically, six components captured 68% of the variance, implying that PCA reduces a dataset to lower dimensions without loss of important information. The transformation makes a dataset more manageable and improves the performance and interpretability of predictive models by focusing on key components rather than the whole set of original variables.

Factor Analysis complemented PCA by giving more insights into the data structure. Only a few factors explain most of the variance, clearly with diminishing returns beyond the first two factors. This is well represented by the scree plot, indicating there is very little added value beyond the second factor in including more factors. This forms a

balance between model complexity and interpretability, thus ensuring that the analysis remains focused only on the most impacting components, hence improving efficiency and accuracy.

Cluster analysis helped to fine-tune the model by classifying data into distinct groups. Silhouette scores showed that either two or three clusters provided both the best separation and cohesion. Higher scores for these configurations indicate well-defined groups. Clustering beyond three groups does not further improve the models, which is shown by their decreasing silhouette scores. This now enables performing more targeted segmentation-based analysis and predictive modelling for operational and environmental conditions.

Thus, all these statistical analyses add to the benefit of the study by enriching knowledge about the data and improving model performance. Correlation analysis warned about feature selection, PCA and factor analysis provided dimensionality reduction and simplification of data, while cluster analysis added actionable results about segmentation. These methods provide a robust set of predictive models—interpretable and effectively capturing complexities in this aviation crash data.

4.3 Predictive models for flight crash risk

One of the major tasks in a host of domains, from finance and cyber security to industrial monitoring, is anomaly detection. It deals with the identification of events or observations that are extremely rarely encountered against most data. In other words, effective anomaly detection could identify potential problems or fraudulent activities which might have been buried in the mass of transactions otherwise. The study is focused on building flight risk prediction based on occurrence code events.

4.3.1 Detailed Framework for Model Training and Evaluation

Building a predictive modelling consist of following steps,

1. Data pre-processing
2. Model training
3. Evaluation

1. Data pre-processing

- a) **Data Loading:** Reading the dataset from a provided CSV file into a Pandas DataFrame was done first. The data was viewed to understand its structure and ready it for further processing.
- b) **Encode Categorical Variables:** Categorical columns in the dataset were identified, and those variables were converted into a numerical format using label encoding. It is a technique through which that categorical data is transformed into a form usable by machine learning algorithms, which usually accept only numerical inputs.
- c) **Feature Selection:** Relevant feature selection for the anomaly detection task was then made. This set contains numerical and previously encoded categorical variables. The choice of these features is relevance-driven, considering the task at hand and how each of them would add to the performance of the model.
- d) **Standardization of Features:** Standardization of the features was done so all of the features were on the same scale and gave equal contribution towards the model. It adjusts the features to have a mean of zero and a standard deviation of one. This step is very important for algorithms sensitive to the scale

of features, for example, K-Nearest Neighbors and Support Vector Machine. This can be assured if the features are standardised; features will all be within the same comparable scale, which means no single feature would come to dominate the fitting of a model.

- e) **Synthetic Data Generation:** Synthetic normal data points were created to form a balanced dataset that will include the simulation of normal instances. The set of data, to be drawn from a normal distribution, would represent the normal class.
- f) **Combining Data and Labelling:** One data set contained synthetic normal data, and the other contained real anomaly data. Labels were assigned for normal and anomaly instances: normal instances were labelled as 0, while anomaly instances were labelled as 1. This combined dataset allows a holistic view of data for training and testing of the models.
- g) **Data Splitting:** Evaluation of developed machine learning models will require a split of this combined dataset into training and test sets. In this case, the data was divided in such a way that 20% of it was used as a test dataset and 80% formed the basis for the training dataset.

2. Model Training

In this research, four different machine learning models are defined, which include K-Nearest Neighbors, Support Vector Machine, Logistic Regression, and Naive Bayes.

- a) **Training:** The training dataset is used to train each model. While training, it learns patterns in the data that help in distinguishing normal from anomalous instances.
- b) **Prediction:** Trained each model and predicted the class labels for the test dataset. These are compared with the true labels to obtain measures of model performance.

3. Evaluation and Visualization

The measure of performance for every model is assessed with the following metrics

- a) **Accuracy:** It is the number of correct predictions divided by the total number of predictions.
- b) **Confusion Matrix:** Confusion matrix is a summary of how many of the predicted classes are true positives, true negatives, false positives, and false negatives.
- c) **Classification Report:** This includes precision, recall, and F1-score for each class, along with macro and weighted averages.
- d) **Results for Each Model:** A plot was generated for every model with the predicted anomalies against actual anomalies. This plot will help in assessing how well each model does in class separation.

A comparison plot for the accuracy of each model has been created. This plot shows, quite clearly and graphically, comparisons of the model performance by pointing out the accuracy for every algorithm.

4.3.2 Performance of the models

Table 5 Overall performance of the models

Algorithms	Accuracy		Precision	Recall	F1-score
KNN	83.56 %	Normal	0.97	0.69	0.86
		Anomaly	0.76	0.98	0.81
SVM	94.01 %	Normal	0.96	0.92	0.94
		Anomaly	0.93	0.96	0.94
Naïve Bayes	65.04 %	Normal	0.82	0.38	0.72
		Anomaly	0.60	0.92	0.52
Logistic Regression	49.67	Normal	0.49	0.53	0.50
		Anomaly	0.50	0.47	0.50

4.3.2.1 K- Nearest Neighbors

The K-Nearest Neighbors algorithm does very well in telling the difference between anomalies and normal cases with an **accuracy of 83.56%**. From the ROC AUC score of 0.9353, this means a high capability of the model to make a distinction between the two classes. However, there exist several trade-offs observed with the confusion matrix and classification report:

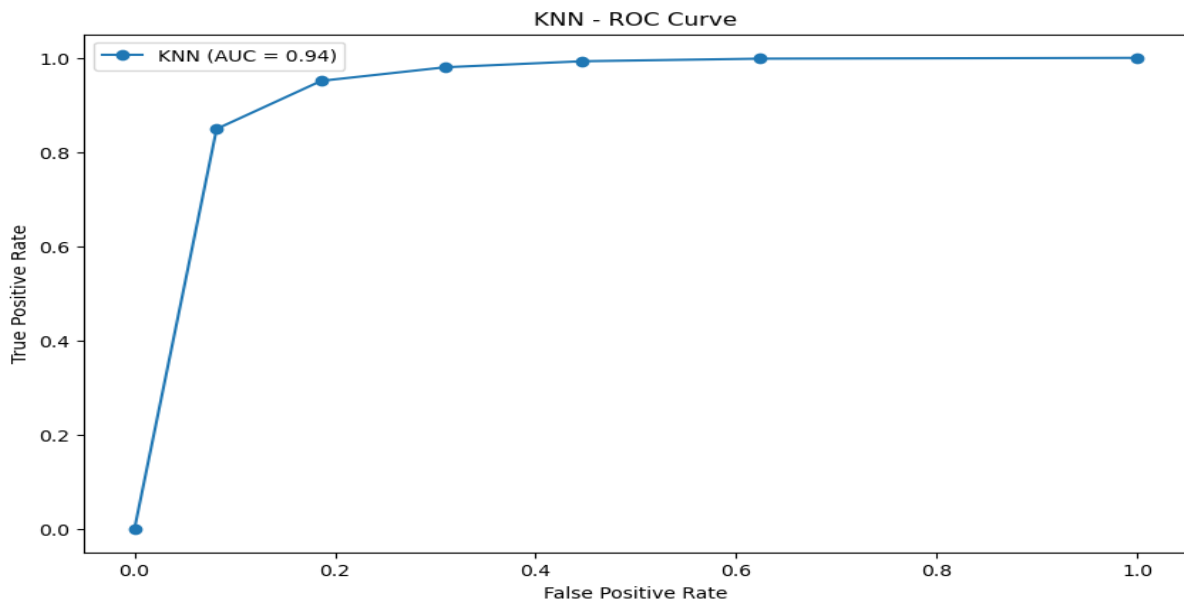


Figure 28 KNN- ROC curve

1. Precision, Recall and F1- Score:

Table 6 KNN results

Class	Precision	Recall	F1- Score
Normal (0)	0.97	0.69	0.86
Anomaly (1)	0.76	0.98	0.81

On Class 0, KNN has high precision at 0.97, thus most of its normal predictions are correct, but it has a lower recall of 0.69, missing a fair number of normals. For Class 1, anomalies, KNN has a precision of 0.76 and a high recall of 0.98, thus it is good at picking up actual anomalies, although at a higher rate of false positives. The score for anomalies in this case is .86, which is better than the one for normal cases. This reflects that the KNN model has quite a balanced approach but with a stronger performance in detecting anomalies.

2. Accuracy:

An accuracy of 83.56% shows that, all things considered, this model runs pretty well; however, there is definitely room for improvement in the classification of normal cases due to this class imbalance in precision and recall.

The high ROC AUC and F1-scores obtained for KNN recommend it in scenarios where the detection of anomalies is more critical than the accurate identification of all normal cases.

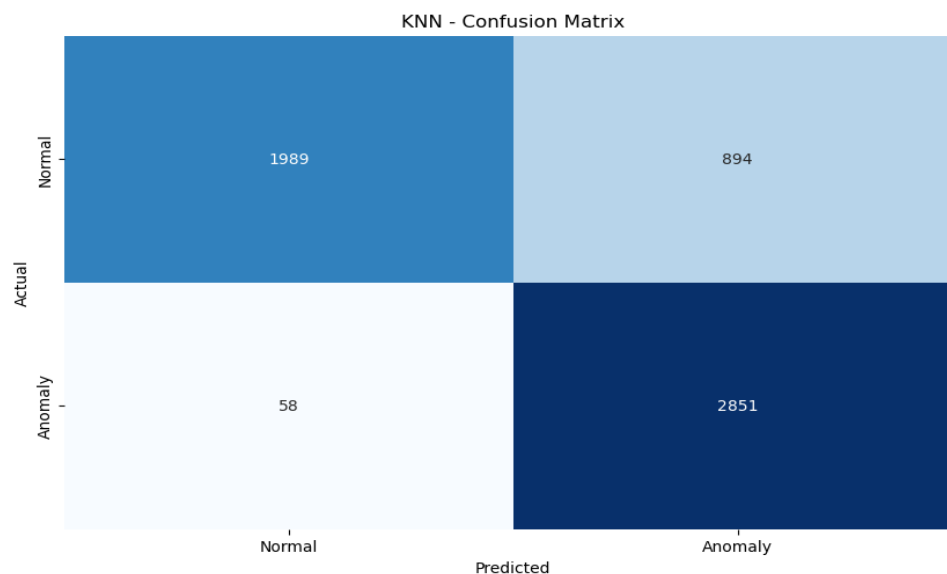


Figure 29 KNN confusion matrix

4.3.2.2 Support Vector Machine

The Support Vector Machine model goes very well with accuracy of 94.01% and an ROC AUC score of 0.9842. As a result, this model performed better in differentiating between normal and anomalous instances.

1. Precision, Recall and F1- Score

Table 7 SVM results

Class	Precision	Recall	F1- Score
Normal (0)	0.96	0.92	0.94
Anomaly (1)	0.93	0.96	0.94

On the normal cases, the SVM model is very accurate. The precision alone is 0.96, with a recall of 0.92, which means low false positives and a high rate of true positives. On anomalies, it gives a precision of 0.93 and a recall of 0.96, which is effective classification with few false positives and a high true positive rate. The balanced F1-score of 0.94 also indicates strong performance overall in detection for both normal and anomaly cases.

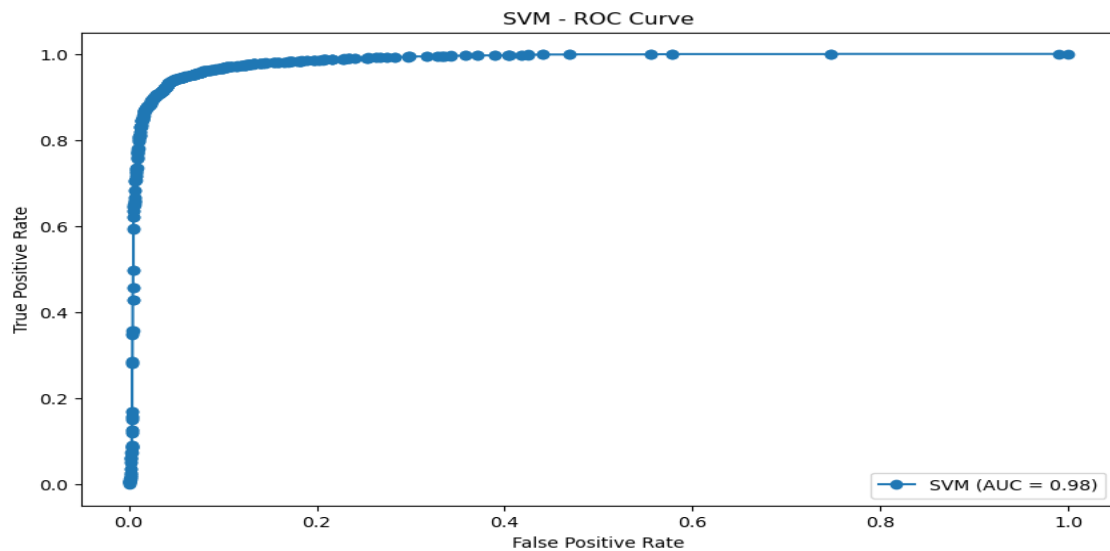


Figure 30 SVM – ROC curve

2. Accuracy:

The high value of 94.01% accuracy confirms that the SVM model is robust and reliable for both precision and recall in detecting normal and anomalous cases. The high ROC AUC score of SVM and the balanced precision and recall in this case render it very excellent in scenarios that require high accuracy and reliability for a classification task.

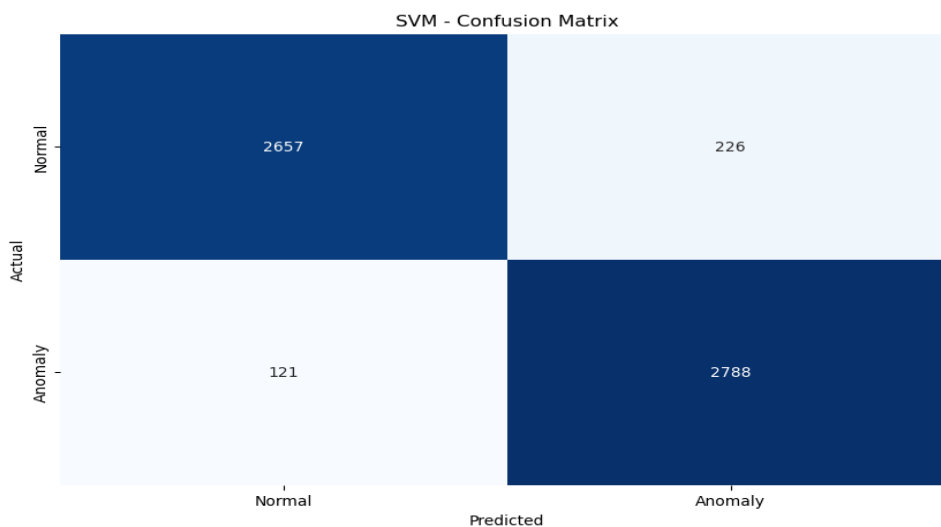


Figure 31 SVM confusion matrix

4.3.2.3 Naïve Bayes

The Naïve Bayes classifier did fairly well, with an accuracy of 65.04 percent and an ROC AUC score of 0.7157, indicating it is better than Logistic Regression but worse than KNN or SVM.

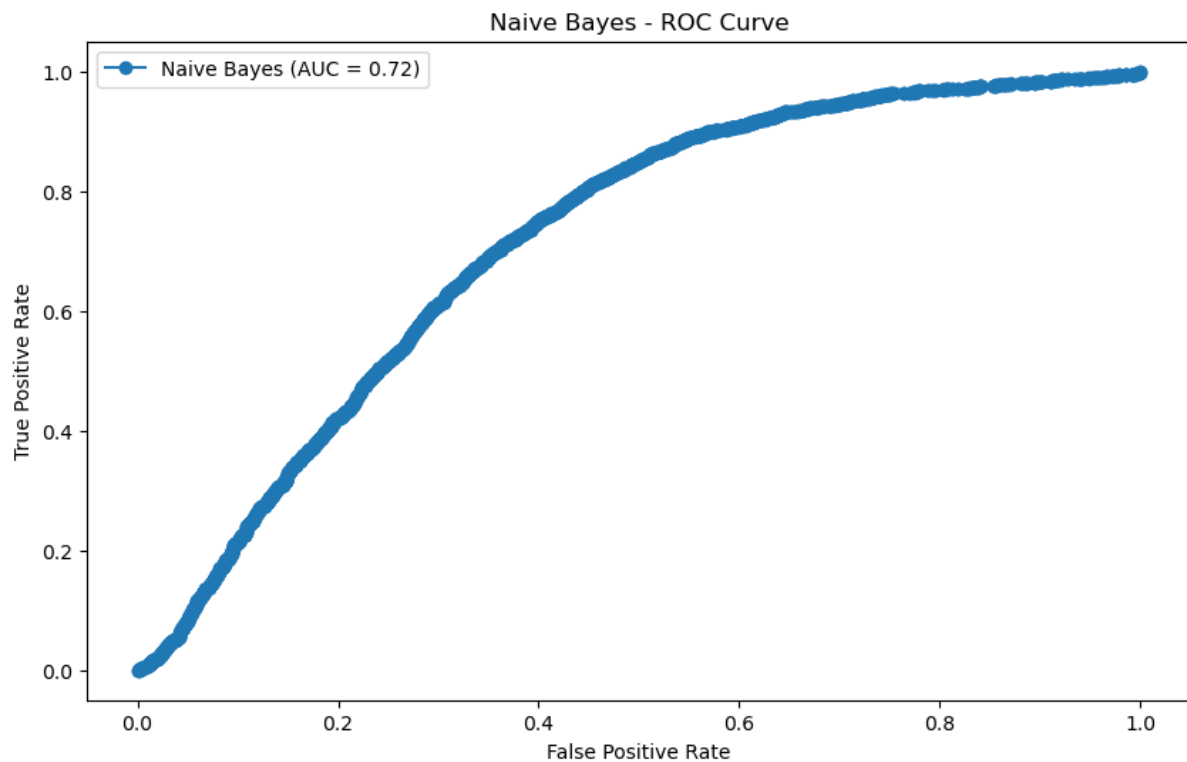


Figure 32 Naïve Bayes ROC curve

1. Precision, Recall and F1-Score

Table 8 Naïve Bayes results

Class	Precision	Recall	F1- Score
Normal (0)	0.82	0.38	0.72
Anomaly (1)	0.60	0.92	0.52

For Class 0, Naive Bayes shows a precision of 0.82 and a recall of 0.38, indicating reasonable accuracy but a tendency to miss many normal cases. For Class 1, the precision is 0.60 with a recall of 0.92, reflecting good detection of anomalies but with more false positives. The F1-score is higher for anomalies (0.72) compared to normal cases (0.52), suggesting the model is more effective in identifying anomalies than normal instances.

3. Accuracy:

65.04% accuracy reflects performance at a medium level but also indicates that it is less reliable as compared to KNN and SVM, particularly for classifying normal cases.

Although Naive Bayes is far superior in detecting anomalies than Logistic Regression, it is by far less effective at this than either KNN or SVM, so in some cases when high recall for anomalies is very critical, Naive Bayes would prove to be a good choice.

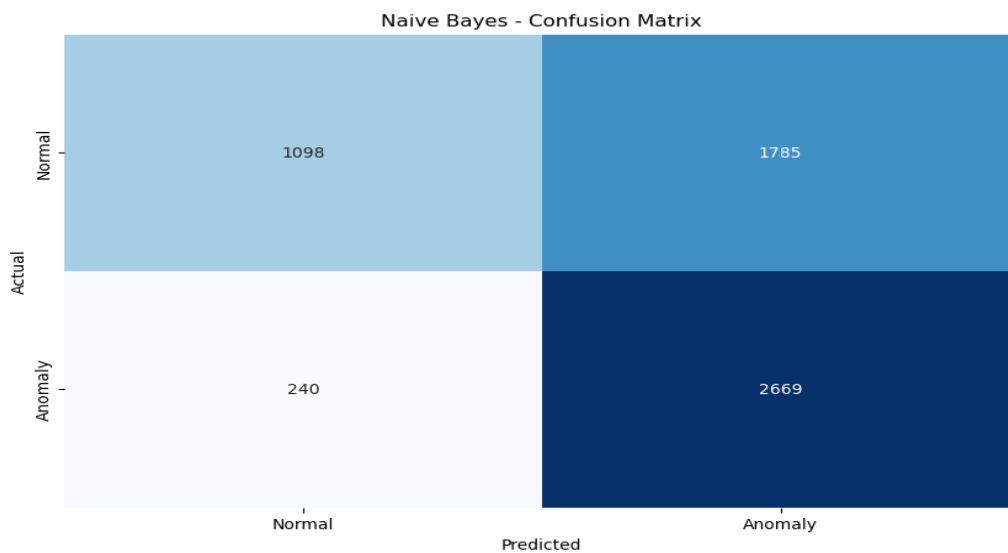


Figure 33 Naïve Bayes Confusion matrix

4.3.2.4 Logistic Regression

Poor performance by Logistic Regression at an accuracy of 49.67% and a ROC AUC score of 0.5020 indicates the struggle of the model to effectively classify between normal and anomalous cases.

1. Precision, Recall and F1-Score:

Table 9 Logistic regression results

Class	Precision	Recall	F1- Score
Normal (0)	0.49	0.53	0.50
Anomaly (1)	0.50	0.47	0.50

Performance metrics for the logistic regression model clearly depict poor performance in classification tasks. For Class 0, which represents the normal cases, precision and recall are 0.49 and 0.53, respectively, indicating very poor identification of normal

cases. For Class 1, representing the anomalies, it is 0.50 in precision and 0.47 in recall, again indicating very poor performance on anomaly detection.

Finally, it is worth mentioning that the F1-score—remaining around 0.5 for both classes—further underlines how the model is imbalanced and fails to find the line separating normal from anomalous cases.

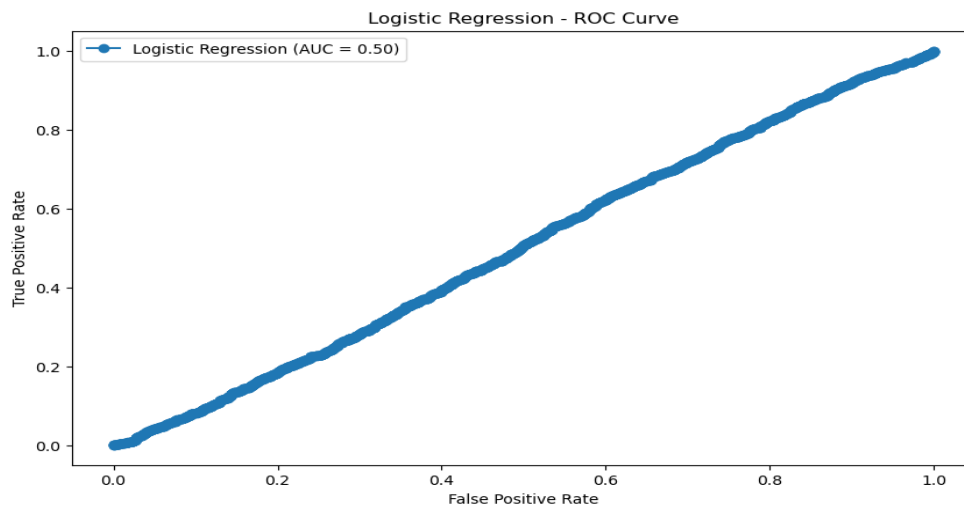


Figure 34 ROC – Logistic regression

2. Accuracy:

The model is not doing any better than random guesses, with an accuracy near 50%. The model, Because of the low ROC AUC and F1-scores, the Logistic Regression does not seem to suit the task. Further tuning or feature engineering may be required.

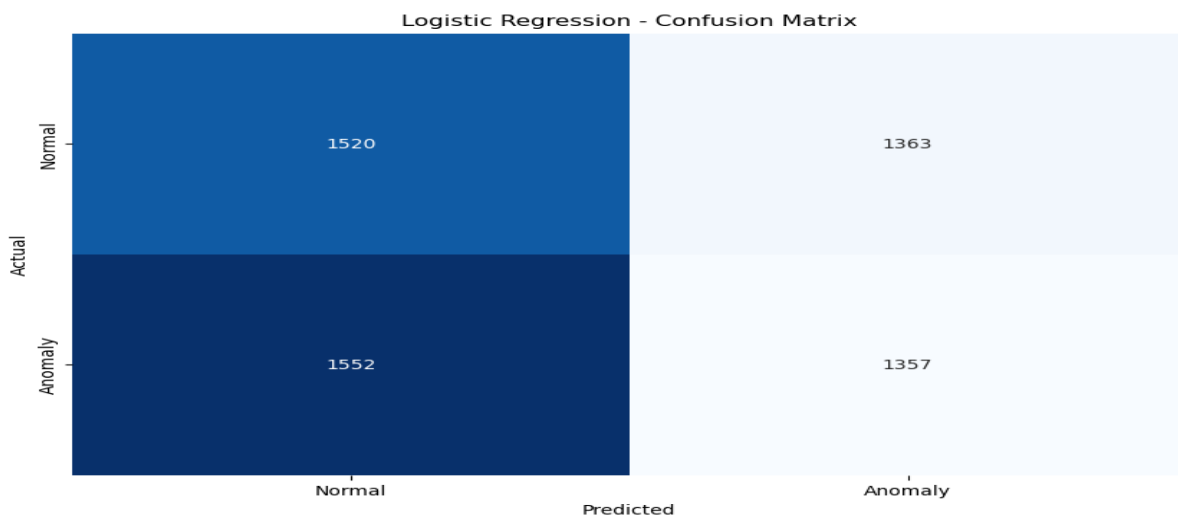


Figure 35 Logistic Regression Confusion matrix

4.3.2.5 Concluded performances:

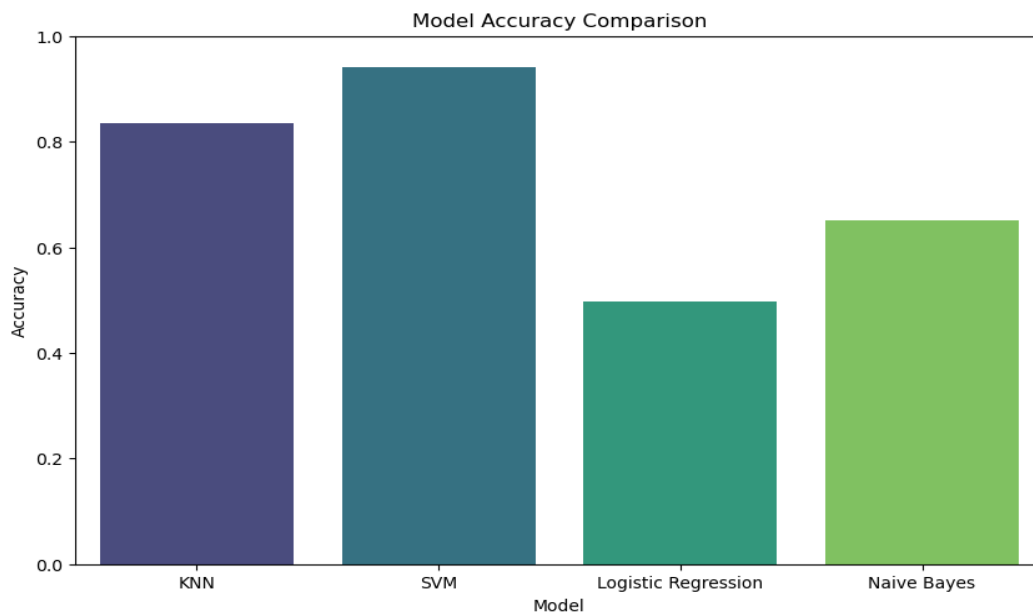


Figure 36 Accuracy comparison between the models

In summary, the evaluated machine learning algorithms for anomaly detection have different strengths and weaknesses as follows:

Support Vector Machine: This model proved to be the most effective among them, showing superior performance in the accurate differentiation between normal and anomalous data. Actually, this is a very strong model due to its high classification accuracy and robustness to class-imbalanced datasets.

K-Nearest Neighbors also performed very well, especially about its precision in finding true anomalies. It, however, had high false positives that will raise false alarms.

Logistic Regression was less accurate and somewhat poor in separating anomalies from normal data. The simplicity in Logistic Regression is its strength but sometimes too simple for complex cases of anomaly detection.

Naive Bayes performed moderately, much better than logistic regression but worse than both SVM and KNN. While it is efficient and straightforward, its effectiveness in this case was limited.

In general, the decision on which algorithm to use would be based on available computational resources and specific requirements of the task regarding accuracy or tolerance for false positives.

5. Project management

In our research, project management as a disciplined approach for planning, executing, and monitoring projects to meet specific objectives effectively while managing scope, time, cost, and quality. It involves important tasks in goal setting, organization of resources, monitoring progress, and staying on track in the project.

5.1 Effective Time Management and Task Insights in Project Execution

The Gantt chart shows time management from the earliest tasks of topic and dataset selection to the latest evaluation and reporting. It highlights that more time has been used for data pre-processing than was initially planned, which means this part is really crucial with regard to data quality and appropriateness. This predictive modelling phase, however, did end up much more efficiently than expected, allowing more time for the complete validation and comparison of various models. With this extra time, it was possible to do a rigorous evaluation on the many different approaches and pick four robust models. Additionally, the consistent updates to the report upon the completion of each task streamlined the process of finalizing the report structure, ensuring a well-organized and coherent final presentation.

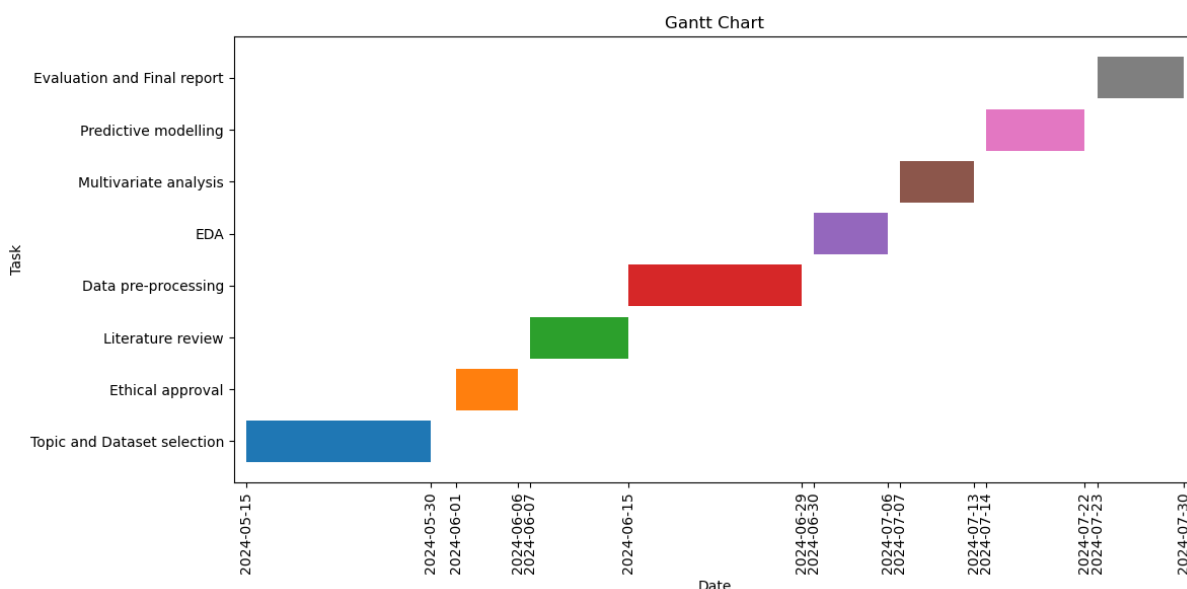


Figure 37 Gantt chart for research plan

5.2 Risk management:

An important aspect for a master's dissertation project is an effective risk management strategy concerning potential failures, data loss, and balancing commitments. The table below outlines key risks and mitigation strategies to ensure project success and the well-being of a person.

Table 10 Risk management and mitigation

Possible risks	Description	Mitigation
Failure of the project	Risk of models failing near the submission deadline.	Established tight validation and testing from the start to enable the identification of problems and fixing them early enough.
Data Loss	Potential loss of work due to laptop failure or theft.	Backed up files regularly to OneDrive for secure keeping and easy recovery.
Balancing Research and Career Development	As a final semester student, challenge of managing project work alongside career development activities like seminars and job interviews.	Employed effective time management strategies and maintained a consistent schedule for project milestones
Health and Mental Well-being	Stress and burnout from intense project work and career preparation.	Integrated frequent breaks, time proportional workload, and stress management practices to maintain sustainability.

Implementing robust risk management strategies ensures that the project progresses smoothly despite potential challenges. By planning for unexpected setbacks, securing data, and balancing academic and personal responsibilities, the project is positioned for successful completion while maintaining a focus on quality and personal well-being.

5.3 Resource management

Resource management in this context involves appropriately using available library resources, laptops, and access to digital libraries for doing research work. This includes accessing materials required on time, managing the technology that will be utilised to analyse data, and seeking academic resources for removal of doubts. Steady progress is also ensured through regular classes and consultations with supervisors. In a job interview, proper management of these resources helps in ensuring that research tasks are executed effectively, challenges are taken care of, and the project is on the right path.

5.4 Quality management

Quality for this project was managed through the regular supervisor meetings that incorporated very pertinent feedback and advice, which kept the work at par with the set goals and standards. Immediately raising doubts and clarifying them with the supervisor averted probable lapses hence maintaining the quality of work. Effective time management kept a well-structured approach to avoid delay risks and ensure the completion of each phase, giving attention to detail. Moreover, the prudent preparation taken for the challenges presented an opportunity to handle the real issues efficiently as they would arise, resulting in an overall successful output of quality research.

5.5 Social, Ethical, Legal and Professional consideration

When developing a predictive model of flight crashes, considerations at the social, ethical, legal, and professional levels are very important. Socially, sensibility is required in handling data about people's privacy and any consequences from such findings on aviation safety or public trust. Ethically, ensuring the confidentiality of sensitive flight data and obtaining appropriate permissions is vital, as is avoiding harm to individuals or groups involved. Legally, there exist personal data and intellectual property rules to be taken into consideration, especially in the case of proprietary aeronautical data. As a professional matter, a good researcher must respect stringent standards of integrity by ensuring that the results are reliable, reproducible, and used for the betterment of safety without damaging the reputation of the field. These aspects are dealt with in detail to guarantee that the research will have a positive contribution toward aviation safety while upholding high ethical and professional standards.

6. Achievements and Critical Evaluation of the Research

6.1 Key takeaways from the Analysis

The analysis gave several key insights into the predictive modelling of flight crash risk using supervised machine learning algorithms. First, it underlined the importance of data pre-processing, particularly feature scaling and encoding that had a very strong impact on model performance. This standardized features to have equal contributions in the training process of the model, hence improving its robustness in making predictions.

6.2 Extent of Analysis in Answering the Research Questions and Objectives

The analysis clearly addressed the primary research questions and objectives. From this, it was well established how efficient the different supervised machine learning algorithms applied in predicting the risk of crashing flights are, and the functions of multivariate analysis techniques in improving model accuracy and interpretability.

a) Effectiveness of Supervised Machine Learning Algorithms:

In this research, several algorithms like K-Nearest Neighbors, Support Vector Machines, Logistic Regression, and Naive Bayes were pitted against each other on flight crash risk prediction. Then, the performance of every algorithm on criteria like accuracy, precision, recall, and F1-score was scored. It showed that although, in many cases, SVM and KNN turned out to be better than others on accuracy and precision,

b) Implications of multivariate analysis techniques:

Multivariate techniques, and PCA particularly, contributed much to the improvement in model performance. Using PCA, the models were improved not only in being more efficient but also interpretable after the dataset was reduced to meaningful components. A scree plot was coupled with an eigenvalue threshold to identify an optimal number of components that would capture as much variance as possible within the retained components. Not only were the models simplified, but it also reduced computational complexity, which reduced the time taken to train the model and improved generalization to new data.

6.3 Addressing the Research Objectives

a) Measuring and Comparing Machine Learning Algorithms:

This study has successfully measured and relatively compared various supervised machine learning algorithms. In this research, it is also shown that each algorithm had some strengths and weaknesses, hence giving a comprehensive understanding of applicability in different scenarios. For example, SVM showed quite good performance for high-dimensional data, while Logistic Regression did for simplicity and interpretability.

b) Impact of Pre-processing Techniques:

Various pre-processing techniques implemented to study their impact on the performance of the model have been dealt with in detail. The importance of feature scaling was very essential for algorithms, especially for SVM and KNN, due to their sensitivity to the scale of the input features. Another important component in this pre-processing step is the encoding of categorical variables, which help the model ensure there is no loss in terms of information while representing the categorical data.

c) Application of Multivariate Analysis Techniques:

Application of multivariate analysis techniques, including PCA and Factor Analysis, was very productive for the models' accuracy and interpretability. Those techniques decreased the noise and redundancy in data by focusing on the most important components and, thus, provided more robust models. It demonstrated that multivariate techniques can handle high-dimensional data and give a clear insight into the underlying patterns and relationships.

6.4 Limitations of the Research

The results of the research were quite good, though with some key limitations, mainly on the dataset. These were mainly on the quality of data and completeness. There are missing values in the dataset that require imputation or exclusion. Either introduces bias or affects the sample size. Further, the dataset may not be big enough to hold complex patterns and relationships that enable highly accurate predictions. Challenges that were then resolved included the presence of outliers and noise, which was sorted out through extensive pre-processing to ensure data quality. In addition, possible limitations of the dataset can also be raised in the sense that it does not represent all scenarios or regions and therefore generalization cannot be established.

7. Conclusion

7.1 Achievement

This was a successful study that covered well the research questions and objectives, giving several valuable insights into the application of machine learning in aviation safety. The findings bring out the role played by data pre-processing and dimensionality reduction techniques in building accurate and interpretable predictive models. However, limitations related to data quality, completeness, and representativeness should be considered. This study has made an in-depth assessment of the performance of various supervised machine learning algorithms and multivariate techniques in the prediction of flight crash risk using aviation safety data. This research serves as the basis for future studies on advanced machine learning and statistical techniques in aviation safety and sets up milestones for more accurate and interpretable predictive models. This research serves to demonstrate how advanced analytical methods contribute to aviation safety by increasing the prediction and prevention of flight crashes, so that people will travel under safer skies.

8. Future Works

While the research has outputted milestones, several areas call for further investigation that needs to be extended with these findings in order to enhance the predictive power of machine learning models in aviation safety.

- **Incorporate More Advanced Algorithms:** In the future, research should focus on applying more sophisticated machine learning algorithms, including deep learning and ensemble methods, neural networks. Such types of methodologies are known to cope elegantly and efficiently with large and complex datasets and may therefore prove to be far more accurate.
- **Expansion in Dataset and improved quality:** One critical limitation of this research was the dataset size and completeness. Thus, for future research, it should aim to be larger and more heterogeneous in datasets to ensure better coverage of most scenarios and regions. This will increase the generalizability and robustness of the models, hence making more reliable predictions across different contexts. In future, works should be channelled into advanced data cleaning and pre-processing methods to handle missing values, outliers, and noise more effectively.

- **Real-time Data Incorporation:** It can be further integrated with real-time data like live weather conditions, aircraft telemetry, and air traffic information to immensely enhance the predictive accuracy of the models. Real-time data will give timely insights that will enable proactive measures to avoid potential flight crashes.

Implementing Recommendations:

All of the above recommendations require certain strategic actions. The aviation authorities and organizations can be collaborated with by sharing larger and more comprehensive datasets. Further, investments in infrastructure to capture and process real-time data will become imperative in this initiative. Developing advanced algorithms which can facilitate real-time risk assessments and undertake better decision-making processes can be done by integrating these algorithms in existing aviation safety systems.

Longitudinal studies on the performance of the predictive models over some time might be very instrumental in giving insight into models' efficiency, effectiveness, and reliability. Such studies enable identification of any drifts or changes in data patterns, hence providing timely updates and adjustments of the models.

9. Student Reflection

This research project has been quite an enriching journey for a young student transitioning from a physics background to a Master's in data science. Coupling this with a passion for aviation and new expertise in data science made the entire project very fulfilling. The prospect of delivering a predictive model for flight crashes presented the quite engaging opportunity to delve into the dynamics of aviation and sophisticated data analysis techniques, therefore quite broadly expanding my knowledge in these fields. This work has sharpened my technical skills and thus boosted my confidence in applying data science to real-world problems in the domain. Navigating complex datasets, mastering advanced methods of analysis, and solving practical problems provided all-round experience. This project has not only solidified my understanding of data science but has further boosted my passion toward aviation, setting a very strong base for my future career. Skills and insights learned here will certainly leverage my abilities in the corporate world, hence making this a very valuable chapter in my academic and professional journey.

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Appendix

This GitHub repository of mine contains entire research's code used for python. It contains jupyter notebook with predictive models, python codes used for EDA and Multivariate analysis, and R language code used for PCA and Clustering. The cleaned dataset after EDA also available in the repository for the verification.

<https://github.com/Ramana-Kulanthaivelu/Predictive-modelling-of-flight-crash-risk>

Scheduled supervisor meetings and insights:

Date	Duration	Venue	Insight
03-06-2024	1 hour	DLC 24	Introduction session and Ethical approval regarding
13-06-2024	30 mins	DLC 24	Literature review regarding
25-06-2024	30 mins	Microsoft Teams	Meetings on current status of the report and interactions about the pre-processing
08-07-2024	30 mins	DLC 24	One to One meeting regarding the current status of the research And Exploratory Data analysis verification
22-07-2024	20 mins	Microsoft Teams	One to One meeting regarding results of Principle component analysis and predictive model along with current report status
31-07-2024	20 mins	DLC 24	Final report verification and feedback session

Regular updates on the status and model were emailed to the supervisor every two weeks. The supervisor provided consistent responses and feedback on each update throughout the process. The final analysis results were included in the report after receiving the supervisor's approval and her verification.

Predictive modelling code:

```

In [1]: # -*- coding: utf-8 -*-
        """
        Created on Tue Jul 30 23:35:49 2024

        @author: Ramana_
        """

        import pandas as pd
        import numpy as np
        from sklearn.preprocessing import StandardScaler, LabelEncoder
        from sklearn.model_selection import train_test_split
        from sklearn.metrics import classification_report, confusion_matrix, accuracy_score, roc_curve, roc_auc_score
        from sklearn.neighbors import KNeighborsClassifier
        from sklearn.svm import SVC
        from sklearn.linear_model import LogisticRegression
        from sklearn.naive_bayes import GaussianNB
        import matplotlib.pyplot as plt
        import seaborn as sns

        # Load the dataset
        data = pd.read_csv("C:/Users/Admin/Desktop/Dissertation/avall (1)/dataset_pca.csv")

        # Define categorical columns
        categorical_columns = ['latitude', 'longitude', 'code', 'Occurrence_Code', 'acft_model', 'light_cond', 'eng_type']

        # Encode categorical variables
        label_encoders = {}
        for col in categorical_columns:
            le = LabelEncoder()
            data[col] = le.fit_transform(data[col].astype(str))
            label_encoders[col] = le

        # Select features for anomaly detection
        features = ['vis_sm', 'crew_age', 'num_eng', 'wx_temp', 'wind_dir_deg', 'wind_vel_kts', 'altimeter'] + categorical_columns
        X = data[features]

        # Standardize the features
        scaler = StandardScaler()
        X_scaled = scaler.fit_transform(X)

        # Generate synthetic normal data points
        normal_data = np.random.normal(loc=0, scale=1, size=(X.shape[0], X.shape[1]))

        # Combine normal data and anomaly data
        X_combined = np.vstack([normal_data, X_scaled])
        y_combined = np.hstack([np.zeros(normal_data.shape[0]), np.ones(X_scaled.shape[0])]) # 0 for normal, 1 for anomalies

        # Split the data into training and test sets
        X_train, X_test, y_train, y_test = train_test_split(X_combined, y_combined, test_size=0.2, random_state=42)

        # Define models
        models = {
            'KNN': KNeighborsClassifier(n_neighbors=5),
            'SVM': SVC(kernel='rbf', gamma='auto'),
            'Logistic Regression': LogisticRegression(random_state=42),
            'Naive Bayes': GaussianNB()
        }

        # Train, predict, and evaluate each model
        results = {}
        for name, model in models.items():
            model.fit(X_train, y_train)
            y_pred = model.predict(X_test)
            accuracy = accuracy_score(y_test, y_pred)
            results[name] = {
                'accuracy': accuracy,
                'confusion_matrix': confusion_matrix(y_test, y_pred),
                'classification_report': classification_report(y_test, y_pred)
            }

        # Plot Confusion Matrix
        plt.figure(figsize=(10, 6))
        cm = results[name]['confusion_matrix']
        sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', cbar=False,
                    xticklabels=['Normal', 'Anomaly'], yticklabels=['Normal', 'Anomaly'])
        plt.title(f'{name} - Confusion Matrix')
        plt.xlabel('Predicted')
        plt.ylabel('Actual')
        plt.show()

        # Plot ROC Curve
        fpr, tpr, _ = roc_curve(y_true, y_prob)
        plt.figure(figsize=(10, 6))
        plt.plot(fpr, tpr, marker='o', label=f'{name} (AUC = {results[name]["roc_auc"]:.2f})')
        plt.xlabel('False Positive Rate')
        plt.ylabel('True Positive Rate')
        plt.title(f'{name} - ROC Curve')

```

```
plt.legend()
plt.show()

# Accuracy Comparison Plot
plt.figure(figsize=(10, 6))
model_names = list(results.keys())
accuracies = [results[name]['accuracy'] for name in model_names]
sns.barplot(x=model_names, y=accuracies, palette='viridis')
plt.title('Model Accuracy Comparison')
plt.xlabel('Model')
plt.ylabel('Accuracy')
plt.ylim(0, 1)
plt.show()

# Print results
for name, result in results.items():
    print(f"{name}:\n")
    print("Confusion Matrix:\n", result['confusion_matrix'])
    print("\nClassification Report:\n", result['classification_report'])
    print(f"Accuracy: {result['accuracy']:.4f}")
    print(f"ROC AUC: {result['roc_auc']:.4f}\n")
```

KNN:

Confusion Matrix:
[[1989 894]
[58 2851]]

Classification Report:

	precision	recall	f1-score	support
0.0	0.97	0.69	0.81	2883
1.0	0.76	0.98	0.86	2909
accuracy			0.84	5792
macro avg	0.87	0.83	0.83	5792
weighted avg	0.87	0.84	0.83	5792

Accuracy: 0.8356
ROC AUC: 0.9353

SVM:

Confusion Matrix:
[[2657 226]
[121 2788]]

Classification Report:

	precision	recall	f1-score	support
0.0	0.96	0.92	0.94	2883
1.0	0.93	0.96	0.94	2909
accuracy			0.94	5792
macro avg	0.94	0.94	0.94	5792
weighted avg	0.94	0.94	0.94	5792

Accuracy: 0.9401
ROC AUC: 0.9842

Logistic Regression:

Confusion Matrix:
[[1520 1363]
[1552 1357]]

Classification Report:

	precision	recall	f1-score	support
0.0	0.49	0.53	0.51	2883
1.0	0.50	0.47	0.48	2909
accuracy			0.50	5792
macro avg	0.50	0.50	0.50	5792
weighted avg	0.50	0.50	0.50	5792

Accuracy: 0.4967
ROC AUC: 0.5020

Naive Bayes:

Confusion Matrix:
[[1098 1785]
[240 2669]]

Classification Report:

	precision	recall	f1-score	support
0.0	0.82	0.38	0.52	2883
1.0	0.60	0.92	0.72	2909
accuracy			0.65	5792
macro avg	0.71	0.65	0.62	5792
weighted avg	0.71	0.65	0.62	5792

Accuracy: 0.6504
ROC AUC: 0.7157

In []:

Ethical Approval

Predictive modeling of flight crash risk - A data-driven approach using Multivariate analysis and Machine learning

P177639



Certificate of Ethical Approval

Applicant: Ramana Kulanthaivelu
Project Title: Predictive modeling of flight crash risk - A data-driven approach using Multivariate analysis and Machine learning

This is to certify that the above named applicant has completed the Coventry University Ethical Approval process and their project has been confirmed and approved as Low Risk

Date of approval: 12 Jun 2024
Project Reference Number: P177639